
Experion™ Pro260 Analysis Kit

Instruction Manual

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BIO-RAD

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Contents

Chapter 1: Experion™ Pro260 Analysis Kit	1
1.1 Product Description	2
1.2 Kit Components	3
1.3 Storage Conditions	3
1.4 Specifications	3
1.5 Additional Requirements	4
Chapter 2: Essential Practices	5
2.1 Storing and Preparing Samples and Reagents	6
2.2 Priming and Loading the Chip	6
2.3 Running the Analysis	7
2.4 General Maintenance	7
2.5 Experion Video Tutorials	7
Chapter 3: Experion Pro260 Assay Procedure	9
3.1 Set Up the Electrophoresis Station	10
3.2 Equilibrate the Kit Reagents	10
3.3 Filter the Gel and Prepare the Gel-Stain Solution	10
3.4 Prepare the Sample Buffers	11
3.4.1 Reducing Conditions	11
3.4.2 Nonreducing Conditions	11
3.5 Prepare the Samples and the Pro260 Ladder	11
3.6 Prime the Chip	12
3.7 Load the Chip	13
3.8 Run the Pro260 Analysis	14
3.9 Clean the Electrodes	15
3.10 Evaluate the Run	16
Chapter 4: Data Analysis	17
4.1 Viewing Data	18
4.1.1 Managing Run Files and Project Folders in the Tree View	18
4.1.2 General Display Controls	19
4.1.3 Electropherogram View	19
4.1.4 Gel View	23
4.1.5 Results and Settings	23

4.2	Changing the Fluorescence Intensity Scale	24
4.3	Using Results and Settings to View and Annotate Data	25
4.4	Comparing Data from Different Runs	26
4.5	Saving, Exporting, and Printing Data	27
4.5.1	Saving Data Files	27
4.5.2	Exporting Data Files to Other Applications	27
4.5.3	Printing Data Files	27
Chapter 5:	Protein Quantitation Methods	29
5.1	Protein Quantitation Methods	30
5.2	Performing Percentage Determination	30
5.3	Performing Relative Quantitation	31
5.4	Performing Absolute Quantitation	32
5.4.1	Absolute Quantitation Using the Upper Marker as Standard	32
5.4.2	Absolute Quantitation Using a User-Defined Internal Standard	34
Chapter 6:	Changing Analysis Settings and Parameters	35
6.1	Designating and Searching for Specific Proteins	36
6.2	Changing Protein Quantitation Parameters	36
6.3	Manually Setting a Marker	36
6.4	Excluding a Peak from Analysis	36
6.5	Changing Peak Finding Parameters	37
6.6	Changing General Settings	37
6.7	Baseline Modification	38
6.8	Turning Analysis Off	39
6.9	Manual Peak Integration	40
Chapter 7:	Troubleshooting	41
7.1	Electrophoresis and Priming Stations	42
7.2	Experion Pro260 Analysis	42
7.3	Contacting Technical Support	46
Appendices		47
	Appendix A: How the Experion System Works	48
	Appendix B: Deep Cleaning Procedure	53
	Appendix C: Chemical Compatibility	54
	Appendix D: Glossary	58
	Appendix E: Bibliography	60
	Appendix F: Ordering Information	61



Experion™ Pro260 Analysis Kit

1.1 Product Description

The Experion Pro260 analysis kit is used for protein analysis with the Experion automated electrophoresis system (Figure 1.1). The Experion system employs LabChip microfluidic technology to automate protein and nucleic acid electrophoresis by integrating separation, detection, and data analysis within a single platform. The Experion automated electrophoresis system uses smaller sample and reagent quantities than standard analysis methods, and it accomplishes analysis in a single 30-minute, automated step.

The Experion Pro260 analysis kit supplies the microfluidic chips, reagents, and instructions required to separate and analyze 10–260 kD proteins under denaturing conditions. The sensitivity of the Experion Pro260 analysis kit is comparable to (though sometimes more sensitive than) colloidal Coomassie Blue staining of SDS-PAGE gels. Each Experion Pro260 chip can analyze up to 10 samples.

For details about how the Experion Pro260 kit analyzes proteins, refer to Appendix A in this manual.

Register your Experion system in order to ensure you receive important updates on software, tech notes, and manuals. Upon installation, a dialog will provide registration instructions.



Fig. 1.1. The Experion system. The system includes the following components: 1) automated electrophoresis station, 2) priming station, 3) vortex station used for nucleic acid analysis only, 4) system operation and data analysis tools (software), and 5) analysis kits, which include the (a) chips and (b) reagents for protein (Pro260 kit), standard-sensitivity RNA (StdSens kit), high-sensitivity RNA (HighSens kit), and DNA (DNA 1K and 12K kits) analyses.

1.2 Kit Components

Table 1.1. Components of the Experion Pro260 analysis kit.

Item	Description	Volume (per Vial)	10-Chip Kit
Pro260 chip	Microfluidic chips used for protein separation		10 chips
Cleaning chip	Chip used for cleaning electrodes		1 chip
Pro260 gel	Proprietary polymeric sieving matrix	520 µl	3 vials
Pro260 stain	Proprietary fluorescent dye	45 µl	1 vial
Sample buffer	Buffer for protein sample preparation; contains a 1.2 kD lower marker and 260 kD upper marker for alignment of samples to the Pro260 ladder	400 µl	1 vial
Pro260 ladder	Protein standard containing 9 purified recombinant proteins of 10–260 kD and optimized for automated electrophoresis on the Experion system	60 µl	1 vial
Spin filters	Used for filtering reagents during sample preparation		3 filters

1.3 Storage Conditions

Table 1.2. Storage conditions.

Item	Storage	Shelf Life
Experion Pro260 reagents	4°C	See expiration date on packaging
Experion Pro260 chips	Ambient	See expiration date on packaging
Gel-stain solution (GS, prepared)	4°C	1 month from filtration (can be refiltered once)
Gel (G, prepared)	4°C	1 month from filtration (can be refiltered once)

1.4 Specifications

Number of sample wells per chip	10
Sample volume required	4 µl
Total run time	~30 min per chip
Protein sizing range	10–260 kD
Limit of detection	2.5 ng/µl carbonic anhydrase (10 ng total) in 1x PBS; similar to colloidal Coomassie Blue G-250 stain
Linear dynamic range	5–2,000 ng/µl (bovine serum albumin in 1x PBS)

1.5 Additional Requirements

Experion automated electrophoresis station

Experion priming station

Microcentrifuge (1,000–10,000 x *g*)

Heating block or water bath set at 95–100°C

Benchtop vortexer

Aluminum foil

Calibrated pipets and narrow-bore tips (for example, VWR #87001-688 or Rainin #L-10F)

Microcentrifuge tubes, 0.5 or 1.5 ml

Deionized water, 0.2 µm-filtered (ReadyPrep™ proteomics grade water, catalog #163-2091)

β-Mercaptoethanol (catalog #161-0710)

Benchtop centrifuge (optional, catalog #166-0612)

Extra spin filter for additional filtration steps (as needed, catalog #700-7254)

Experion electrode cleaner (catalog #700-7252)

2

Essential Practices

2.1 Storing and Preparing Samples and Reagents

Store all Experion™ Pro260 reagents at 4°C when not in use. Do not store reagents at room temperature for >2 hr, as this will shorten their shelf life.

Before use, allow all kit reagents to equilibrate to room temperature (~15–20 min). Once thawed, gently vortex all kit reagents before use. Before opening the tubes, quickly centrifuge them to collect solution to the bottoms of tubes.

If the Pro260 gel has frozen, discard it.

Protect the stain, sample buffer, and gel-stain solution (GS) from light: store these solutions in a dark place and keep them covered with foil when using them.

The Pro260 stain contains DMSO, which is hygroscopic. Cap tightly.

Use GS and filtered gel (G) for up to 1 month. After that, GS and G can be refiltered once again.

Prepare samples in either reducing or nonreducing sample buffer, but always prepare Pro260 ladder in reducing sample buffer.

Do not use coated or treated pipet tips or microcentrifuge tubes (for example, siliconized polypropylene) for preparation of kit reagents or samples. Use of treated tips or tubes may cause separation artifacts.

Use 0.2 µm-filtered or ReadyPrep™ proteomics grade water. Do not use autoclaved water for sample or reagent preparation.

If possible, use sample protein concentrations (before preparation with the Pro260 sample buffer) that are near the middle of the linear dynamic range.

2.2 Priming and Loading the Chip

To avoid contamination, wear gloves and handle chips by the edges. Never touch the glass portions of the chip.

Load the chip on a benchtop or in the priming station. Never load a chip in the electrophoresis station.

Avoid sources of dust and other contaminants when preparing samples and loading the chip. Foreign particles in reagents, samples, or the wells of the chip interfere with separation. Remove chips from their packaging immediately before use.

It may be easier to load the chip on a white background. Tilt the chip to look for bubbles.

Use narrow-bore pipet tips for loading the chip (for example, VWR #87001-688 or Rainin #L-10F).

To avoid introducing air bubbles, do the following (for more help with chip loading, refer to the Experion Training Video in the Experion software Help section under **Contents and Index > Contents >**

Appendices > Technical Videos):

- Insert the pipet tip all the way to the bottom of the chip well when dispensing liquids (this reduces the possibility of trapping air)
- Hold the tip vertically, perpendicular to the chip surface. Holding the tip at an angle may trap air bubbles at the bottom of the well
- When expelling liquid, dispense slowly and only to the first stop on the pipet. Using the second stop introduces air and bubbles into the liquid. Reverse pipetting is acceptable

Dislodge bubbles at the bottom of a well with a clean pipet tip, or remove the solution and load it again.

Use a primed and loaded chip within 5 min of loading. When chips are not used within this time, reagents may evaporate, leading to poor results or a chip performance error.

Fill all the chip wells when running an analysis. Use blank samples (prepared with water instead of sample) or replicates if necessary. All 16 electrode pins must be in contact with liquid; otherwise, an IV (current voltage) check failure error will occur.

2.3 Running the Analysis

Place the electrophoresis station on a stable surface, where it will not be subjected to vibrations or other movement, and away from direct sunlight and all other potential sources of extreme heat.

Power on the electrophoresis station before launching Experion software.

The first time that the Experion electrophoresis station is used, confirm that communication has been established between the software and electrophoresis station before preparing the reagents.

Do not open the lid of the electrophoresis station during a run. The run will abort if the lid is opened.

2.4 General Maintenance

For recommendations on general instrument maintenance, refer to the Experion system manual (bulletin 10001312).

Clean the electrodes after each run (routine cleaning). Cleaning maintains the instrument in optimum condition and prevents buildup and cross-contamination of reagents and samples.

Perform the deep cleaning procedure described in Appendix B to clean the electrodes:

- Prior to first use of the Experion electrophoresis station
- Whenever contamination is suspected or visible (for example, salt deposits or other precipitates) on the electrodes
- Whenever a chip has been left in the electrophoresis station for an extended period of time (for example, overnight)

Never store the cleaning chip inside the electrophoresis station. Store the empty cleaning chip covered to keep the wells clean. A new cleaning chip is included with every box of chips.

2.5 Experion Video Tutorials

For additional information, view the video tutorials available online at www.bio-rad.com:

North America: Home > Life Science Research > Support > Tutorials > Electrophoresis and Blotting > Experion System Training

Other: Home > Life Science Research > Electrophoresis > Automated Electrophoresis > Experion Training Videos



Experion™ Pro260 Assay Procedure

For an abbreviated version of this protocol, refer to the Quick Guide provided with the kit.

3.1 Set Up the Electrophoresis Station

1. If needed, perform a deep cleaning of the electrodes (see Appendix B for instructions).
2. Power on the computer and then power on the Experion electrophoresis station by pushing the green button in the center of the front panel. The steady green LED above the button indicates that the unit is on.
3. Launch Experion software. If the instrument and computer are communicating properly:
 - A green dot and the last 4 digits of the instrument serial number appear in the lower right corner of the software screen
 - The electrophoresis station icon appears in the upper left corner

When there is no connection, these indicators are absent and a grayed-out instrument icon appears in the upper left corner of the software screen.

3.2 Equilibrate the Kit Reagents

1. Set a heating block or water bath to 95–100°C. You will use this heating block to denature the samples and Pro260 ladder later in the protocol.
2. Equilibrate the following kit reagents to room temperature for ~15–20 min:
 - Pro260 stain (blue cap)
 - Pro260 sample buffer (yellow cap)
 - Pro260 ladder (red cap)
 - 2 tubes Pro260 gel (green cap)
3. Vortex the contents of each tube and briefly centrifuge the solutions to the bottoms of the tubes. Make sure the Pro260 stain solution (blue cap) is thawed before proceeding.

If the gel-stain solution (GS) and filtered gel (G) were prepared previously, equilibrate them to room temperature. Use the GS and G within 1 month of preparation. After 1 month, refilter them before use. Keep both the G and GS at room temperature and covered until ready for use.

3.3 Filter the Gel and Prepare the Gel-Stain Solution

1. Prepare the GS by adding 20 µl Pro260 stain (blue cap) to a tube of Pro260 gel (green cap, 520 µl). Vortex the GS for 10 sec at the highest setting and then spin it down briefly in a microcentrifuge.
2. Transfer the GS to a spin filter, and transfer the contents of the other Pro260 gel (green cap) into another spin filter. Label and date the tubes.
3. Centrifuge both spin filters for 5 min at 10,000 × g. Inspect the tubes to ensure all of the gel passed through the filters, and then discard the filters. Cover the GS with foil.

3.4 Prepare the Sample Buffers

Separate protein samples under either reducing or nonreducing conditions, but always use reducing conditions for separation of the Pro260 ladder. Reduced and nonreduced samples can be run on the same chip. Prepare fresh sample buffer daily.

3.4.1 Reducing Conditions

For each chip, combine 1 μ l β -mercaptoethanol and 30 μ l sample buffer (yellow cap). Vortex and spin down briefly. Protect the sample buffer from light.

At pH <7, reducing agents such as β -mercaptoethanol and DTT are less effective; use 5–7.5 mM tributylphosphine (TBP) or tris(2-carboxyethyl)phosphine (TCEP) instead. Otherwise, neutralize the buffer or use buffers of higher pH before preparing samples for Experion runs.

3.4.2 Nonreducing Conditions

Prepare two stocks of sample buffer: one reducing (for the Pro260 ladder) and one nonreducing (for the protein samples).

1. For each chip, transfer 30 μ l sample buffer (yellow cap) to 2 microcentrifuge tubes.
2. Add 1 μ l β -mercaptoethanol to one tube (reducing sample buffer) and 1 μ l deionized water to the other (nonreducing sample buffer). This generates enough nonreducing sample buffer for 10 samples. Vortex the tubes and cover them with foil to protect them from light.

3.5 Prepare the Samples and the Pro260 Ladder

Each chip can analyze up to 10 protein samples. All wells of a chip must be filled for the electrophoresis station to operate properly.

The total protein concentration, including any added user-defined internal standards, in all starting samples should be toward the middle of the linear dynamic range of the assay, if possible.

1. Prepare the Pro260 ladder by combining 4 μ l Pro260 ladder (red cap) and 2 μ l sample buffer with β -mercaptoethanol (reducing sample buffer) in a microcentrifuge tube.
2. Prepare the samples by combining 4 μ l sample and 2 μ l sample buffer (Section 3.4) in a microcentrifuge tube.
3. Vortex all tubes briefly and spin down in a microcentrifuge for a few sec.
4. Heat the samples and Pro260 ladder at 95–100°C for 3–5 min. Spin down the tubes in a microcentrifuge for a few sec.
5. Add 84 μ l deionized water to each tube and vortex briefly to mix. Do not modify this step to adjust sample concentration.

Both the diluted samples and the Pro260 ladder are stable for several hours when stored at room temperature and protected from light.

3.6 Prime the Chip

Start the run within 5 min of priming and loading the chip. For help with chip loading, refer to the Experion Training Video: Chip Loading, available in the Experion software Help section under **Contents and Index > Contents > Appendices > Technical Videos**.

1. Pipet 12 μ l GS into the top right well of the chip (highlighted and labeled GS, gel priming well) (Figure 3.1). Insert the pipet tip vertically and to the bottom of the well when dispensing. Dispense slowly to the first stop on the pipet, and do not expel air at the end of the pipetting step.
2. On the priming station, set the pressure setting to **B** and the time setting to **3**, as specified by the alphanumeric code on the chip (Figure 3.1).

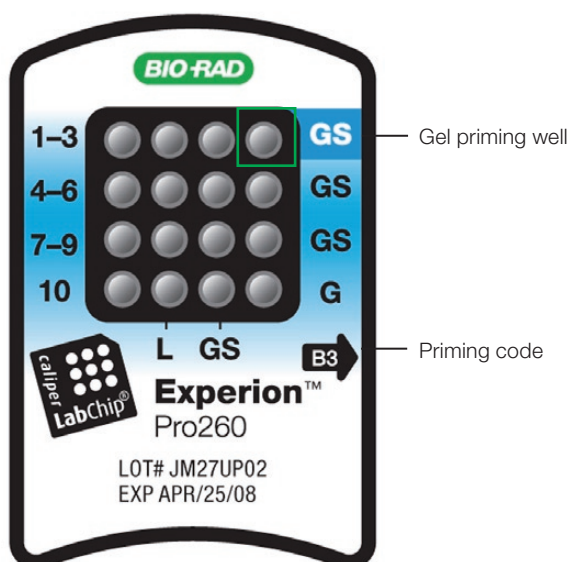


Fig. 3.1. Experion Pro260 chip. The locations of the gel priming well (GS, highlighted) and alphanumeric priming code are indicated.

3. Open the Experion priming station and place the chip on the chip platform, matching the arrow on the chip with the alignment arrow on the chip platform. A post on the chip prevents insertion in the wrong position. Do not force the chip into position.
4. Close the priming station by pressing down on the lid. The lid should snap closed.
5. Press **Start**. A “Priming” message appears on the screen of the priming station, and the timer counts down. Priming requires approximately 60 sec. Do not open the priming station during countdown.
6. An audible signal and “Ready” message indicate that priming is complete. Open the priming station and remove the chip. If the lid sticks, press down on it while pressing down on the release lever.
7. Turn the chip over and inspect the microchannels for bubbles or evidence of incomplete priming. If the chip is primed properly, the microchannels will be difficult to see (it may be helpful to compare a primed chip to a new, unused chip). If you detect a problem, such as a bubble or incomplete priming, prime a new chip.
8. Place the chip on a clean surface for loading.

Bubbles forced into microchannels during priming take the shape of the microchannel and are elongated, not round.

3.7 Load the Chip

1. Using a pipet, remove and discard any remaining GS from the gel priming well. Pipet 12 μ l GS into all 4 wells labeled **GS** (including the gel priming well, Figure 3.2).
2. Pipet 12 μ l filtered gel (G) into the well labeled **G** (Figure 3.2).
3. Pipet 6 μ l of each diluted sample into sample wells **1–10**.
4. Pipet 6 μ l diluted Pro260 ladder into the ladder well labeled **L** (Figure 3.2). Use the Pro260 ladder within 8 hr of preparation. Every chip must have Pro260 ladder loaded into the ladder well labeled **L**.

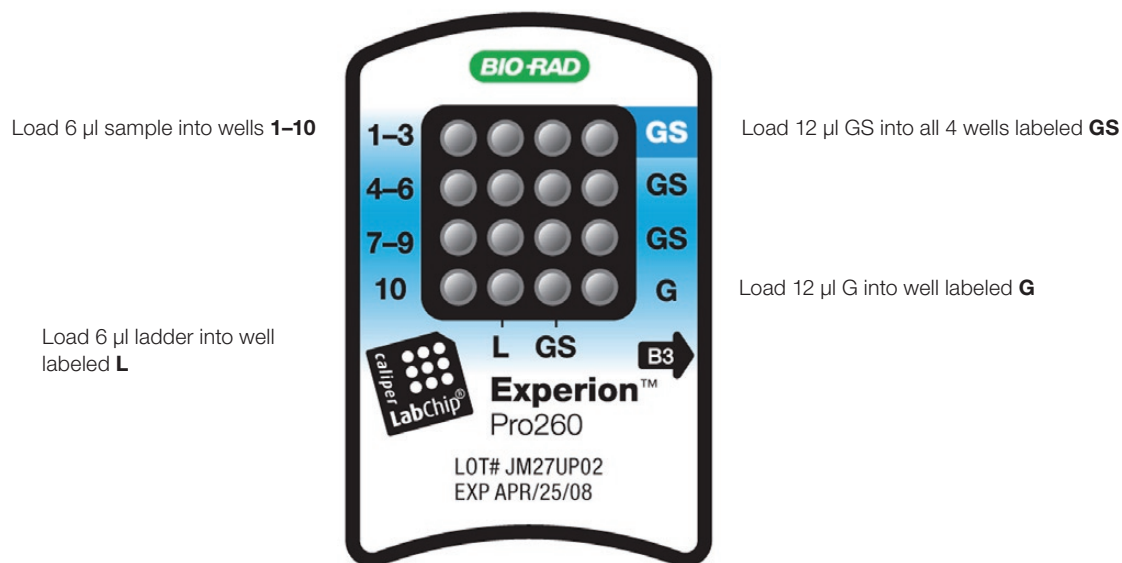


Fig. 3.2. Experion Pro260 chip. Wells for loading GS, G, samples, and ladder are indicated.

5. Inspect all wells for bubbles by holding the chip above a light-colored background and looking through the wells (Figure 3.3). Dislodge any bubbles at the bottom of a well with a clean pipet tip or by removing and reloading the solution.
6. Place the loaded chip into the Experion electrophoresis station and start the run within 5 min.

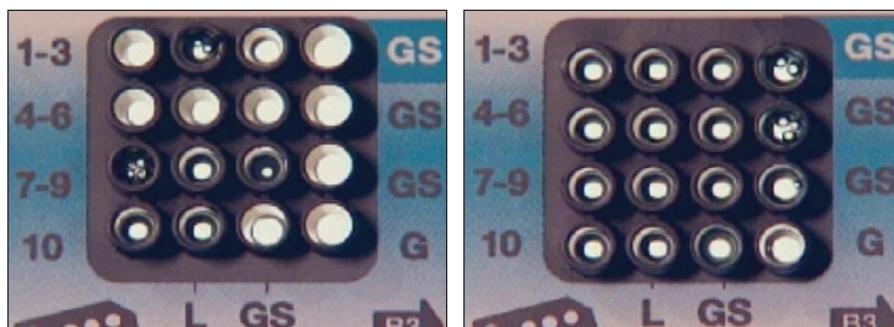


Fig. 3.3. Bubble formation during loading of Experion Pro260 chips. Surface bubbles do not generally cause problems during a run, but bubbles at the bottoms of wells must be removed. **Left**, bubbles trapped at the bottom of wells. The GS and G wells and sample wells 1, 3, and 4–6 contain no solution. Wells 8, 10, and L are filled properly and have no bubbles, but large bubbles have formed at the bottoms of wells 7 and 9 (note the difference in the diameter of the light-colored circles in wells 8 and 9). **Right**, bubbles have formed at the surface of the three GS wells on the right side of the chip; the rest of the wells have no bubbles.

3.8 Run the Pro260 Analysis

1. Open the lid of the electrophoresis station by pulling the release latch. Place the primed and loaded chip on the chip platform and close the lid.
2. In the Experion software toolbar, click **New Run** . In the **New Run** screen (Figure 3.4), from the **Assay** pull-down list, select **Protein > Protein 260**.
3. Either select a project folder for the run from the **Project** pull-down list or create a new project folder by entering a name in the **Project** field or by selecting **File > Project > New**. The project folder appears in the project tree.
4. Enter a name for the run in the **Run Prefix** field and click **Start Run** .

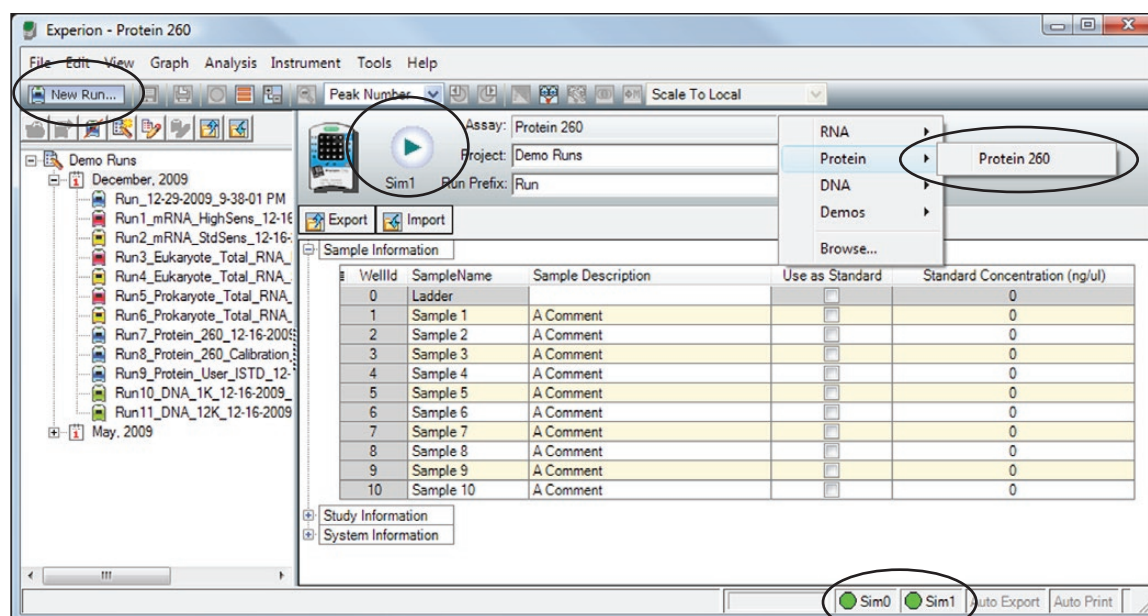


Fig. 3.4. Details of the New Run screen. The green dot in the lower right corner indicates that communication between the electrophoresis station and Experion software has been established.

5. In the **New Run** dialog (Figure 3.5), select the number of samples to be analyzed. Though all wells are filled, the Experion system stops the analysis when it reaches the number of samples entered.

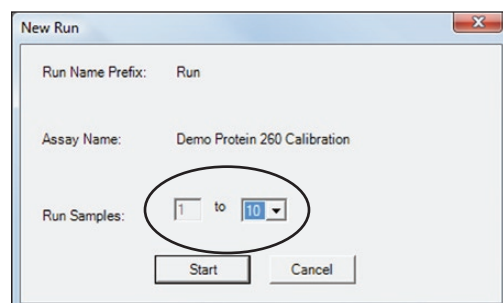


Fig. 3.5. New Run dialog. The Experion system stops analysis when it reaches the number of samples entered.

6. Click **Start**. The green LED in the center of the front panel on the electrophoresis station blinks, and the system performs a number of checks: it confirms that a chip has been inserted, that all wells contain liquid, that electrical connections are made, etc. A calibration counter marks the progress of these calibrations at the upper right of the screen.

Do not open the lid of the Experion electrophoresis station until the run is complete. The lid does not lock. Opening the lid aborts the run.

An “IV Check Error” message indicates the system cannot make electrical contact in one of the wells. This often means there is a bubble at the bottom of the well. Abort the run, and check the chip for bubbles or empty wells. Refill the affected well(s), and start the run again.

7. During separation, the sample name is highlighted in the project tree and the electropherogram trace, and virtual gel bands appear in real time:
 - The electropherogram of the sample being separated appears in the electropherogram view
 - The lane corresponding to that sample is outlined in pink and has a dark background

To display the electropherogram of another sample, click on either the sample name in the project tree or on a lane in the virtual gel.

8. When analysis is complete (after ~30 min), the instrument beeps and a window opens indicating the end of the run. Select **OK** and remove the chip from the chip platform.
9. Clean the electrodes using deionized water within 30 min of each Pro260 chip run.

3.9 Clean the Electrodes

1. Fill a cleaning chip with 800 µl deionized water (0.2 µm-filtered). Gently tap the side of the cleaning chip to remove any trapped bubbles from the wells.
2. Place the cleaning chip on the chip platform in the electrophoresis station, close the lid, and leave it closed for 1 min.

Never store the cleaning chip inside the electrophoresis station. Store the empty cleaning chip covered to keep the wells clean. A cleaning chip is included with each box of chips.

3. Open the lid, remove the cleaning chip, and allow the electrodes to dry for 1 min. Close the lid.
4. Replace the water in the cleaning chip after use to avoid contamination. For storage, remove the water from the cleaning chip and store the chip in a clean location.

3.10 Evaluate the Run

When a run is complete, evaluate the run and the analysis of the data by Experion software.

1. Ensure all lanes (ladder and samples) are visible in the virtual gel. The markers (indicated by pink triangles) should be visible in and aligned across all lanes. If the marker peaks are not properly assigned, you may need to include or exclude peaks, or manually set the marker (see Section 6.3).
2. Evaluate the separation of the Pro260 ladder. To display the ladder electropherogram, click the ladder well in the project tree, or click on the lane labeled **L** in the virtual gel. The electropherogram should resemble the one shown in Figure 3.6 and should have the following features (if your ladder does not have these features, see Chapter 7, Troubleshooting for more information):
 - Two marker peaks and a set of system peaks
 - Eight Pro260 ladder peaks between the system peaks and upper marker
 - Flat baseline
 - Marker peaks at least 20 fluorescence units above the baseline

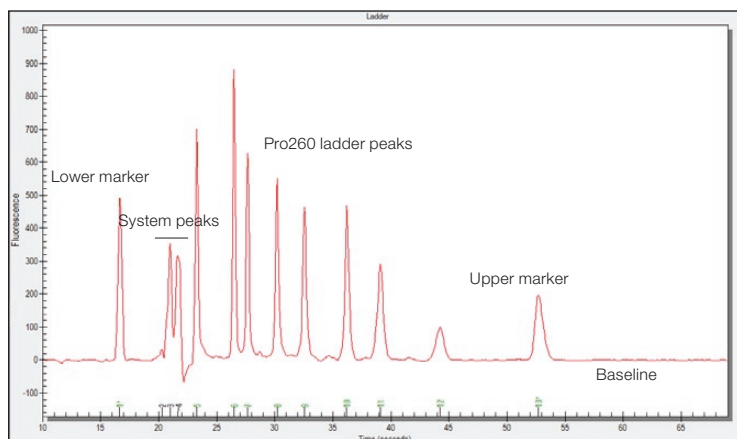


Fig. 3.6. Separation of the Pro260 ladder. Note the flat baseline and well-resolved peaks. All identified peaks are numbered, and the lower and upper markers are indicated by green asterisks (*). The 260 kD protein in the Pro260 ladder is labeled as the upper marker in the Results table.

3. Examine the separation of at least one sample. Click on the sample name in the project tree or on the lane in the virtual gel to view the electropherogram, which should have the following features:
 - Two well-resolved marker peaks and a set of system peaks
 - Sample peaks located between the system peaks and upper marker
 - Flat baseline

If the upper and lower marker peaks are not properly assigned, you may need to include or exclude peaks, or manually set the marker (see Section 6.3).

4. Evaluate the data analysis performed by Experion software (see Chapter 4, Data Analysis). If necessary, change the analysis settings and parameters — including protein quantitation methods — by following the instructions in Chapters 5 and 6.

4

Data Analysis

This chapter outlines the basic steps and software features used to view and analyze data. For a more detailed description of Experion™ software and its various functions, refer to the software Help menu. For information on how to customize analysis parameters, refer to Chapter 6, Changing Analysis Settings and Parameters.

4.1 Viewing Data

The main window of the Experion software user interface contains menus and toolbars, a tree view, and three data views: the electropherogram view, gel view, and results and settings tabs. (Figure 4.1).

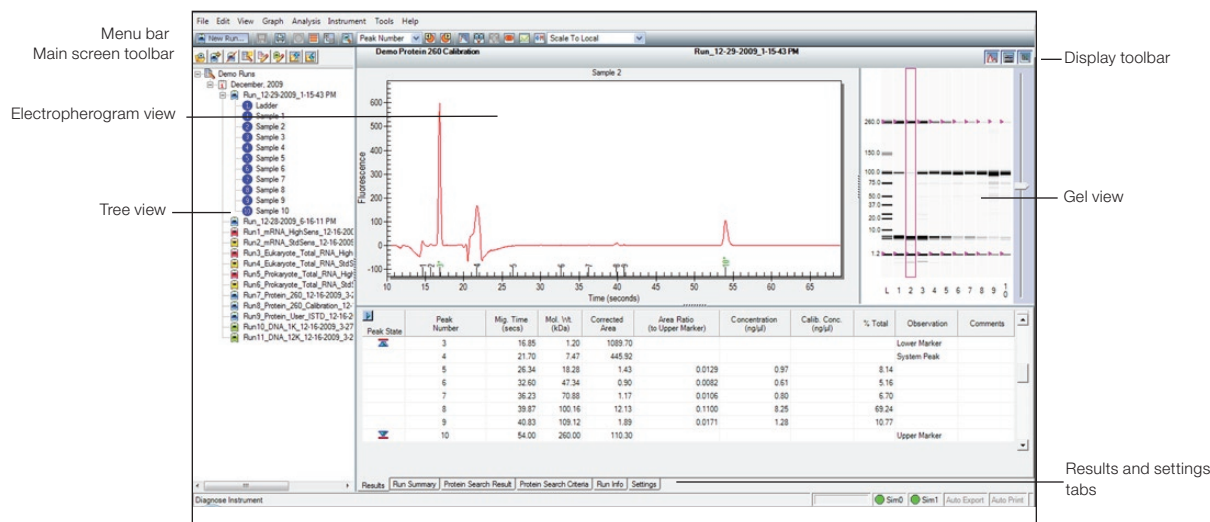


Fig. 4.1. Main window of Experion software. The toolbars, menus, and data views are indicated.

4.1.1 Managing Run Files and Project Folders in the Tree View

The tree view (Figure 4.2) displays the project tree, a hierarchical tree of all projects and the data files they contain. Use the project tree and the toolbar above it to:

- Open run files and select samples for viewing
- Create project folders
- Export and import one or more run files
- Edit names of run files and project folders
- Move runs into project folders
- Delete run files and project folders

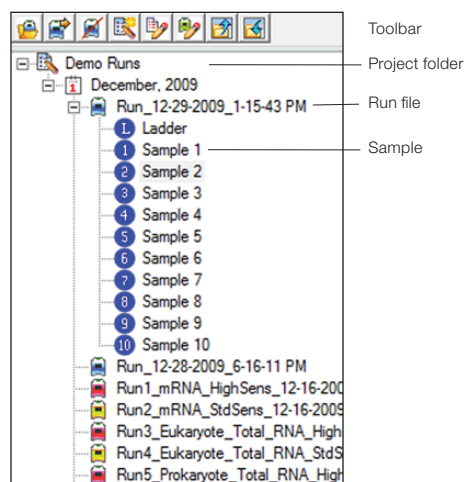


Fig. 4.2. Tree view details. The project folder name, sample well label, and run file prefix can be customized.

To open a run file, double-click on it. The run file opens, displaying the electropherogram and Results tab for the Pro260 ladder, and the virtual gel for the entire run.

To view the data from a sample, click on the sample name or select the corresponding lane in the gel view.

To edit a run file or project folder name, right-click on it and select **Rename Project** or **Rename Run**. Type in the new name.

To import, export, move, or delete files and folders, click the icons in the toolbar or use the context menu accessed by right-clicking on the items in the tree view. For more details, refer to the Experion software Help file.

4.1.2 General Display Controls

Data in the electropherogram view, gel view, and results and settings tabs are linked. Selecting a sample or peak in one area automatically opens or highlights that sample/peak in the other two areas.

To adjust the relative sizes of the three data views and the tree view, click on and drag the border of the view to its new location.

To hide or show one of the three data views, click on the corresponding icon in the display toolbar (located in the upper-right corner of the display, just above the gel view (Figure 4.1).

To zoom in on (expand the view of) a portion of an electropherogram or virtual gel, click on the corner of the area that you would like to enlarge and mouse over to the opposite corner. Double-click anywhere in the electropherogram view to return to the previous view.

4.1.3 Electropherogram View

Experion software plots fluorescence intensity versus migration time to generate an electropherogram for each sample. In the electropherogram, peaks identified by Experion software are numbered (Figure 4.3). Peaks generated by the markers and used for normalization (for example, the lower and upper markers) are numbered in green and labeled with an asterisk.

To view the calculated size of a peak or other area of an electropherogram, place the cursor over the peak/area (Figure 4.3). To display the sizes of all the peaks in the electropherogram, select **Molecular Weight** in the **Select Peak Information** pull-down list (main screen toolbar).

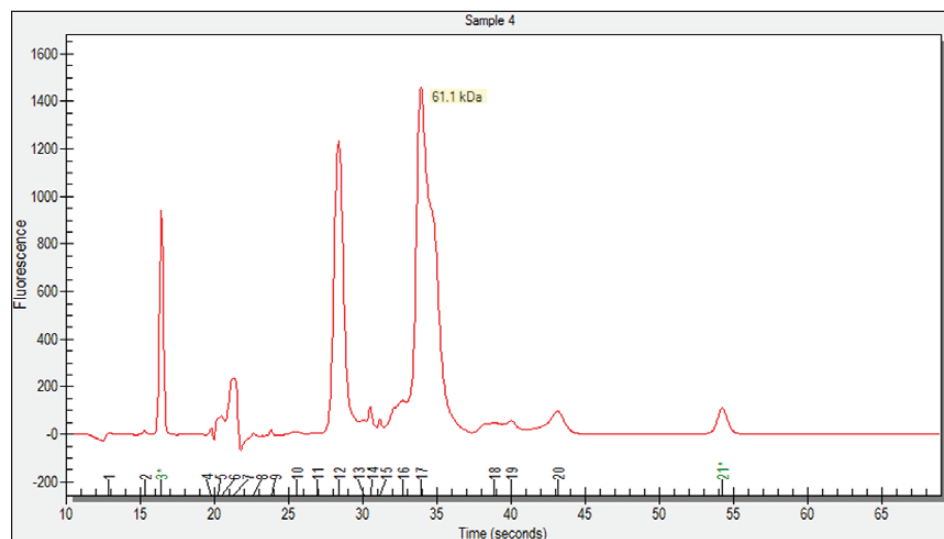


Fig. 4.3. Example of an electropherogram. All identified peaks are numbered, and the lower and upper markers are indicated by a green asterisk (*). Place the cursor over a peak to reveal its size.

Double- or right-click on a peak to select it. An inverted black arrow appears above the peak in the electropherogram, and a pink arrow appears above the corresponding band in the virtual gel. The peak number and corresponding data also appear highlighted in the Results table.

Display electropherograms in three different ways:

- One at a time (single-well view)
- All at once in separate windows (all-wells view)
- As superimposed images (overlays)

Single-Well View

This is the default view, which shows the electropherogram data from a single well (Figure 4.4). To return to this display:

1. Click on the sample in the project tree or on the corresponding lane in the virtual gel.
2. In the main screen toolbar, click **View Single Well**  or select **View > Single Well**.
3. The electropherogram appears. The corresponding lane in the virtual gel is outlined in pink.

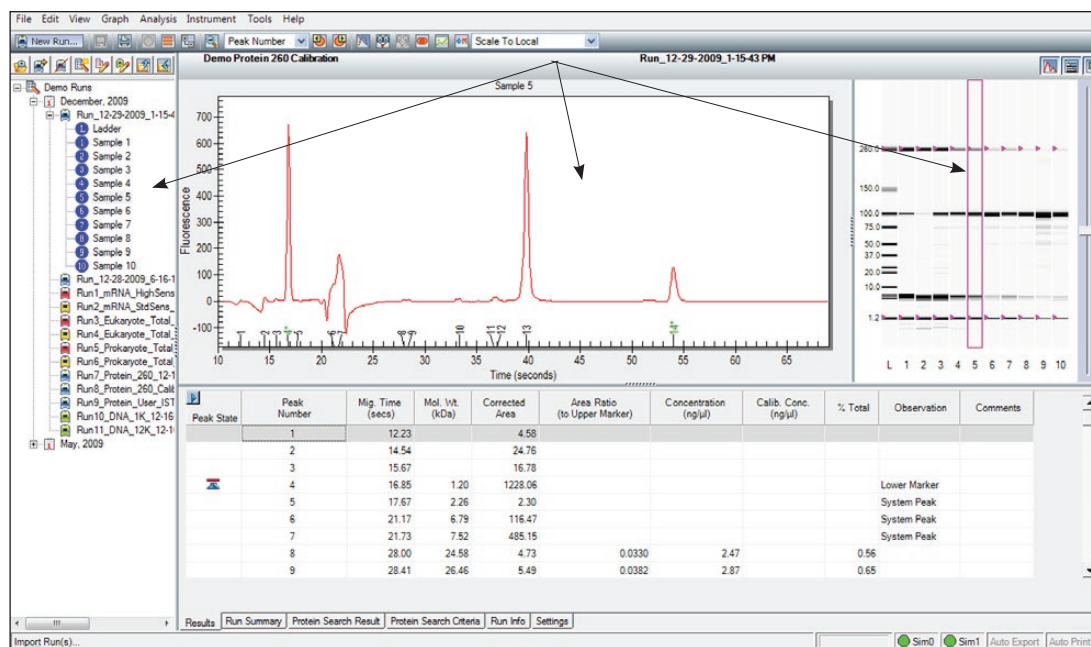


Fig. 4.4. Single-well view. As indicated by the arrows, a single electropherogram appears, the name of the sample is highlighted in the project tree, and the corresponding lane in the virtual gel is outlined.

All-Wells View

To display all electropherograms simultaneously (Figure 4.5):

1. In the main screen toolbar, click **View All Wells**  or select **View > All Wells**.
2. Electropherograms for each sample populate the electropherogram view (Figure 4.5). Use the scroll bar to scroll through the electropherograms, or minimize the results and settings table to accommodate the multiple electropherograms.
3. To select and expand a particular electropherogram, double-click on it. The data appear in a single-well view.

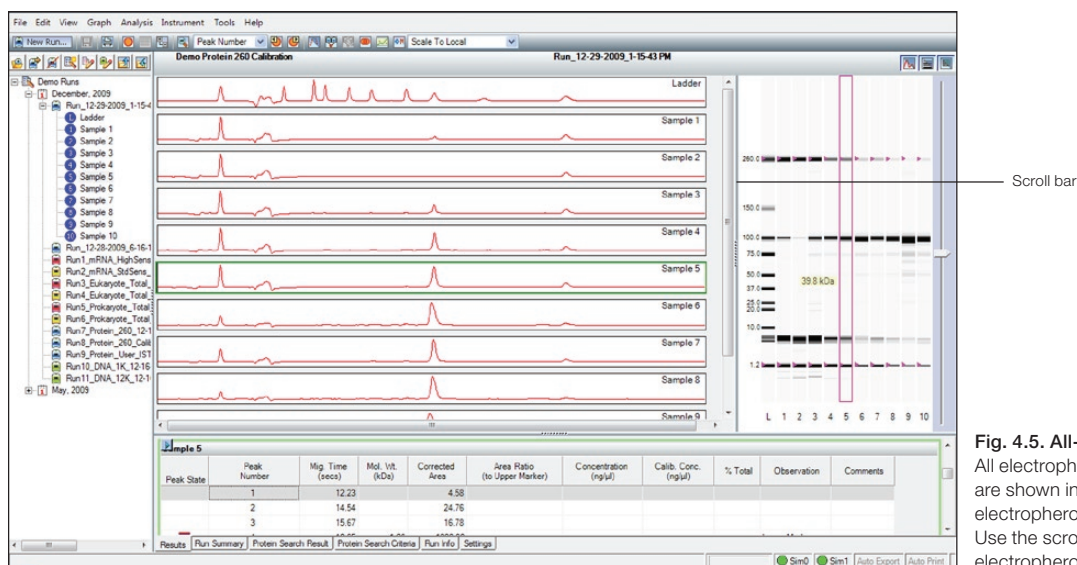


Fig. 4.5. All-wells view. All electropherograms are shown in the electropherogram view. Use the scroll bar to view electropherograms.

Electropherogram Overlays

Superimpose (overlay) multiple electropherograms to facilitate direct comparison among profiles. To overlay a subset of electropherograms (Figure 4.6), use one of the following options:

- Move into the single-well view and, in the main screen toolbar, click **Start Overlay**. Select wells (lanes) in the virtual gel. Each gel lane that is displayed in the electropherogram is outlined. To end the overlay, click **End Overlay**.
- Select the lanes in the virtual gel while holding the **Ctrl** key (for noncontiguous, individual lanes) or the **Shift** key (for lanes next to the original lane selected). To exit, click on a lane in the virtual gel or on a sample name in the project tree
- Select **Analysis > Start Overlay**. Select the lanes to overlay. To end the overlay, select **Analysis > End Overlay**

Electropherograms appear superimposed in a single window, each in a different color. (The colors cannot be changed.) The sample name appears at the top of the electropherogram, in a color that corresponds to the trace. The fluorescence scale of the overlay electropherogram is based on the sample that is selected first; therefore, select the sample with the highest peaks first.

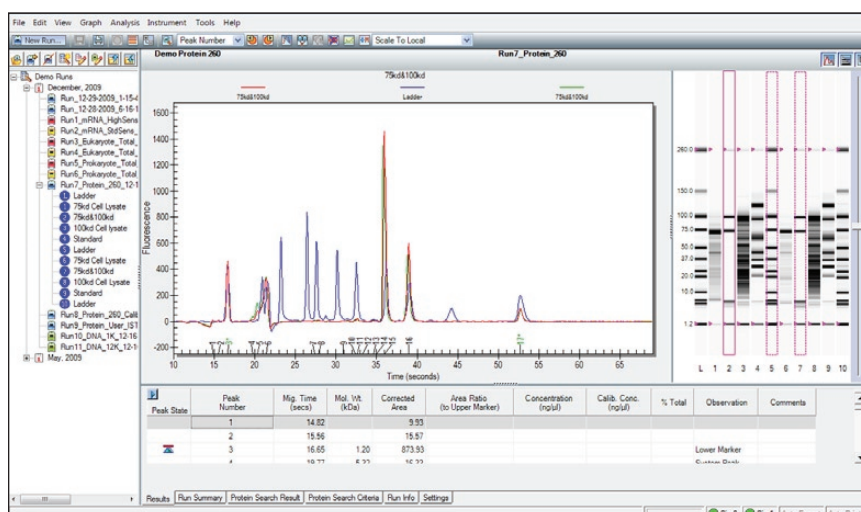


Fig. 4.6. Electropherogram overlay. Electropherograms are shown in different colors, and the corresponding lanes in the virtual gel are outlined. In this example, lanes 2, 5, and 7 are superimposed using the overlay feature.

Peak Labels

Experion software automatically labels detected peaks with numbers for easy identification, and the upper and lower markers used for normalization are indicated in green and with an asterisk (*).

To change the type of label displayed, select **Graph > Peak Info** (or use the pull-down list in the main screen toolbar) and select one of the following options:

- **No Peak Info** — displays no peak information
- **Peak Number** — uses sequential numbers for peak identification (default selection)
- **Peak Time** — uses peak migration time (min:sec) for peak labels
- **Peak Height** — uses peak height in units of fluorescence intensity for peak labels
- **Peak Cor. Area** — uses calculated (corrected) peak area for peak labels
- **Molecular Weight** — uses calculated mass (molecular weight, kD) for peak labels
- **Peak Concentration** — uses calculated relative concentration (ng/μl) for peak labels

Electropherogram Tags (Annotations)

To annotate a peak in the electropherogram, right-click on the peak and select **Add Tag** (Figure 4.7). A tag is automatically generated with the peak information selected (for example, peak number).

To edit a tag, right-click on it, select **Edit Tag**, and change the information. Click **OK**.

To annotate a region in the electropherogram, right-click in the region and select **Add Tag**. In the **Edit Tag** dialog, enter the annotation and click **OK**.

To move a tag, click and drag the label to the desired place in the electropherogram.

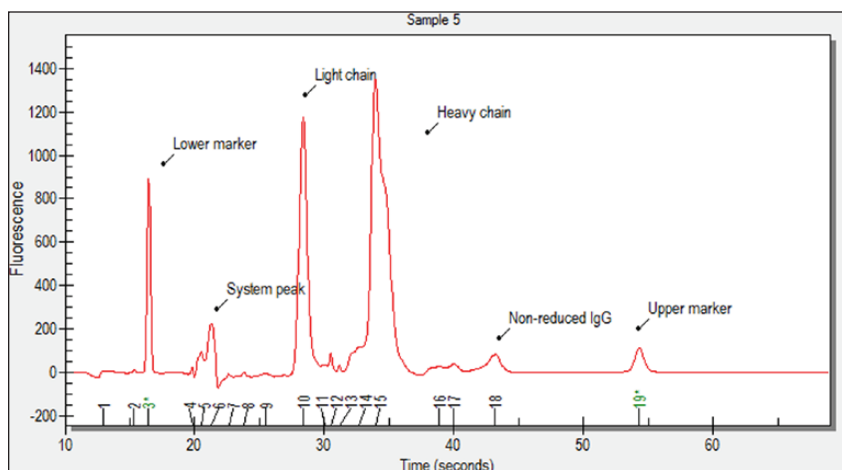


Fig. 4.7. Example of tagged (annotated) peaks in an electropherogram.

4.1.4 Gel View

Experion software converts electropherogram data into densitometric bands, which appear in the virtual gel (Figure 4.8). Each lane of the virtual gel corresponds to a different sample, and all samples from a chip are shown in a single gel view. When a sample is selected for view as an electropherogram, the corresponding lane in the virtual gel is outlined in pink.

Place the cursor over an area in the virtual gel to view its estimated size in kilodaltons ("kDa"). If a band corresponds to a peak identified by Experion software, view the peak number and calculated size, concentration, and percentage of total protein by placing the cursor over the band (Figure 4.8).

Right-click on a band to select it. A pink arrow appears above the selected band in the virtual gel, and an inverted black arrow appears above the corresponding peak in the electropherogram. The peak number and corresponding data also appear highlighted in the Results table. Only one peak can be selected at a time.

To select a different color scheme, choose **View > Gel Color** and select a color from the list. By default, the gel view displays bands as black signals on a white background.

To change the contrast of the bands, use the sliding cursor (Figure 4.8). Changing the contrast does not change the data, but it may improve visualization of faint bands in the virtual gel.

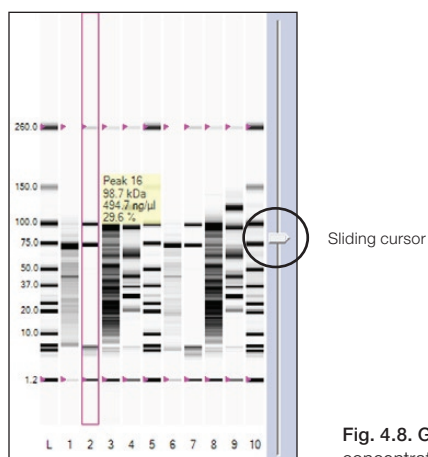


Fig. 4.8. Gel view. Place the cursor over a band in the gel view to display the peak number, size, concentration, and % of total protein. Move the sliding cursor to adjust band intensity.

4.1.5 Results and Settings

The six results and settings tabs (Figure 4.9) present options for viewing, customizing, and analyzing the separation data (see Section 4.3, Using Results and Settings to View and Annotate Data).

Peak State	Peak Number	Mig. Time (secs)	Mol. Wt. (kDa)	Corrected Area	Area Ratio (to Upper Marker)	Concentration (ng/ul)	Calib. Conc. (ng/ul)	% Total	Observation	Comments
	1	14.63		38.75						
	2	15.69		12.15						
	3	16.85	1.20	1089.70						Lower Marker
	4	21.70	7.47	445.92						System Peak
	5	26.34	18.28	1.43	0.0129	0.97		8.14		
	6	32.60	47.34	0.90	0.0082	0.61		5.16		
	7	36.23	70.88	1.17	0.0106	0.80		6.70		
	8	39.87	100.16	12.13	0.1100	8.25	48.95	69.24		Standard Protein
	9	40.83	109.12	1.89	0.0171	1.28		10.77		
	10	54.00	260.00	110.30						Upper Marker
Sample Peaks Found 5										
Total Concentration 11.9 ng/ul										
Results Run Summary Protein Search Result Protein Search Criteria Run Info Settings										

Fig. 4.9. Results and settings. Shown is the Results table for a single sample.

4.2 Changing the Fluorescence Intensity Scale

To display more or less detail in the electropherogram traces and lanes in the virtual gel, use the options available under the **Graph** menu, which determine the scale used for the y-axis (fluorescence intensity) of electropherograms and define the relative scale intensity of the lanes in the virtual gel (Figure 4.10):

- **Scale To Selected** — uses the fluorescence value of the tallest peak in a selected electropherogram to set the scale of the y-axis for all the other samples. Use this option when comparing (overlying) a sample with a high-intensity peak and one with a less intense peak
- **Scale To Global** — uses the fluorescence value of the tallest peak from all samples to set the scale of the y-axis for all the samples. Use this option to adjust the virtual gel to be most similar to an SDS-PAGE separation
- **Scale To Local** (default) — uses the fluorescence value of the tallest peak in each electropherogram to set the scale of the y-axis for that electropherogram: each sample is scaled separately. This is the best way to view protein electropherograms and sample details
- **Scale To Selected Peak** — uses the fluorescence value of the selected peak to set the scale of the y-axis for all samples

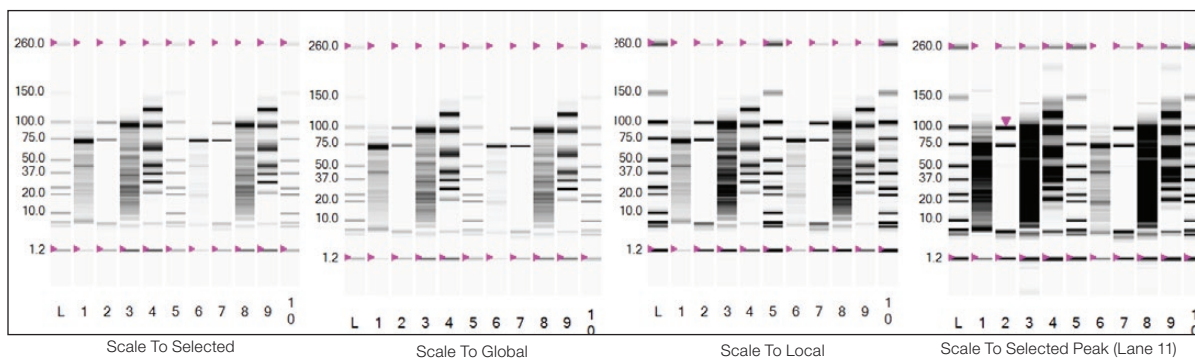


Fig. 4.10. Options for modifying the fluorescence intensity scale. Examples of how each option affects the display of a virtual gel are shown.

4.3 Using Results and Settings to View and Annotate Data

The following options appear under the results and settings tabs:

- **Results** — summarizes analysis data for all detected peaks. The data in this table can be exported (see Section 4.5.2, Exporting Data Files to Other Applications). To view the results of a particular sample, click the electropherogram in the all-wells view or lane in the gel view, or select that sample from the project tree. Alternatively, use the arrows in the main screen toolbar  to navigate through the samples. See below for instructions on how to customize the types of data shown under this tab
- **Run Summary** — displays a run summary for all wells; right-click to export the data
- **Protein Search Result** — summarizes user-defined known proteins: lists the samples that contain a peak that fits with the search of size, or molecular mass as defined under the **Protein Search Criteria** tab and performs statistical calculations for the size, concentration, and percentage of total protein for each peak identified
- **Protein Search Criteria** — used to compare statistics for specific proteins in the samples (see Section 6.1, Designating and Searching for Specific Proteins for information on how to designate and then search for known proteins)
- **Run Info** — used to enter sample names and comments after a run is finished (see below for instructions)
- **Settings** — used mostly for troubleshooting, but can be used to edit the parameters used to identify peaks (see Section 6.5, Changing Peak Finding Parameters, or the Experion software Help file for more details about these settings)

To customize the types of data shown under the **Results** tab:

1. Right-click on any of the column headers in the Results tab (or click the arrow to the left of the column header). The **Column Selector** dialog opens.
2. Use the arrow keys to show, hide, or change the order of display of any of the available headers. For detailed explanations of each setting, refer to the software Help file.
3. Click **OK** to save the changes, or click **Cancel** to close the dialog without saving the changes.

Under the Results tab, total concentration data appear at the bottom of the table. This calculation represents the total protein concentration of the original sample. See Bio-Rad bulletins 5299, 5784, and 5423 to learn more about comparing these calculations to data gathered by other methods.

To enter sample names and descriptions:

1. Click the **Run Info** tab and then click on the plus sign (+) next to **Sample Information**.
2. Click on any active cell, enter the sample names and descriptions, and click **Apply**. Click **File/Save** to save the changes. The sample name appears above its electropherogram.

To enter other information about the experiment:

1. In the **Run Info** tab, click the plus sign (+) next to the **Study Information** and/or **System Information** area(s).
2. Double-click in any blank field, enter the information, and click **Apply**. Click **File/Save**.

4.4 Comparing Data from Different Runs

Experion software allows you to overlay runs from different chips (same assay type), enabling direct comparison of electropherograms and virtual gels from multiple chips. You can save and edit this new virtual chip, but you will not be able to make any new calculations. All calculations in the Results table carry over from the original runs.

In security mode, all run comparisons are saved and cannot be deleted.

1. Click **Create a New Compare Run**  or select **Analysis > New Compare Run**. In the **Compare Runs Setting** dialog, enter a name for the comparison in the **Compare Name** field.
2. Select the project in which you wish to store the comparison from the **Target Project** pull-down list.
3. Select the assay from the **Assay Type** pull-down list and click **Next**. The **Compare Runs** dialog opens, displaying only runs of the same assay type.
4. Double-click to expand the run files you want to compare, or select a run file and click **List Samples**.
5. Click the run file or sample(s) and use the arrows to move them to and from the **Compare** field. To select several lanes at a time, click the sample names while holding the **Ctrl** or **Shift** key. To open a window displaying the selected samples in a gel view, click **Show Gel Lanes**.

Experion software accepts up to 40 samples for comparison (the default is 30). To change the number of wells used, select **Tools > Options**. Under the **Advanced** tab, change the number under **Visible Compare Runs Gel** lanes.

Only one ladder can be used per comparison. If a ladder is not selected, the ladder used in the first run file is used as the ladder for the entire virtual chip. If an additional ladder is inserted in the virtual chip, it is treated as a sample.

6. Select **Realign Data** to align the results of all the runs and click **OK**. The separations appear together in one window as a single experiment. In the Results table, the sizes and concentrations for proteins in each sample are unchanged; the original sizes and concentrations are shown.
7. To change the lanes used in the run comparison:
 - a. Open a run comparison by double-clicking the file name in the tree view and then clicking **Edit Compare Run**  or selecting **Analysis > Edit Compare Run**.
 - b. Select the samples you want to add or delete, and use the arrows to add or remove them from the comparison.
 - c. To view the samples within a run, double-click the run file or select it and click **List Samples**.
8. Click **OK** to process the changes and view the results.

4.5 Saving, Exporting, and Printing Data

4.5.1 Saving Data Files

Data are automatically saved to a run file. Names for run files are given according to the selection made in the **Run Files** tab of the **Options** dialog.

To designate a new name for a file, select **File > Save As** and enter the desired name.

To view the selections or make changes in the automatic naming of files, select **Tools > Options > Run Files**. The following options appear:

- **Create Run name by combining** — select the options to be added to the file name. Options include the prefix (enter the desired characters), assay class (adds the assay type to the file name), instrument name, date, and time
- **Run file folder** — choose between default (saves files to the directory C:\Program Files\BioRad Laboratories\Experion Software\Data) and custom (enables navigation and selection of another folder)
- **Create daily folders** — select to enable creation of daily subfolders by date. When this option is selected, the software stores all run files from the same date in a single folder

4.5.2 Exporting Data Files to Other Applications

To export data to another program (for example, Excel software), select **File > Export Data**. In the **Export** dialog, select the desired options. For a complete description of all export options, refer to the Experion software Help file.

To export files automatically after a run is completed:

1. Select **Tools > Options** and click the **Advanced** tab.
2. Select **Auto Export** and then click **Settings**.
3. Select the options for **Auto Export** and click **Apply**.

To quickly copy and paste electropherograms, virtual gels or lanes in the gels, or data tables into another application, use the options under the **Edit** pull-down list.

4.5.3 Printing Data Files

You can print both graphic and tabulated data from a run.

To choose the print options, select **File > Print** and select the desired options in the **Print** dialog. For a complete description of all printing options, refer to the Experion software Help file (search term “printing data”).

To print files automatically after a run is completed:

1. Select **Tools > Options** and click the **Advanced** tab.
2. Select **Auto Print** and then click **Settings**.
3. Choose the desired options in the **Experion Print** dialog and click **Apply**.

To save or print the data as a PDF file, select **File > Print** and select **Print PDF**.



Protein Quantitation Methods

5.1 Protein Quantitation Methods

Experion™ software enables protein quantitation by several approaches. The selection of approaches depends on the type of experiment and desired level of accuracy (Table 5.1). Refer to Appendix A and bulletin 5784 for details about the different quantitation methods.

Table 5.1. Protein quantitation methods used by Experion software. Absolute quantitation methods provide greater accuracy than relative quantitation methods. The accuracy of percentage determination is protein dependent.

Quantitation Method	Output	Calibration Method and Internal Standard Used			
		Single-Point		Calibration Curve	
		Upper Marker	User-Defined	Upper Marker	User-Defined
Percentage Determination	% Total	No internal standard required			
Concentration Determination					
Relative quantitation	ng/μl	•	—	—	—
		—	•	—	—
Absolute quantitation	ng/μl	—	—	•	—
		—	—	—	•

5.2 Performing Percentage Determination

Experion software automatically calculates a percentage of total protein (% total) value for each detected peak and lists the results in the **% Total** column of the **Results** table (Figure 5.1). Click on the peak of interest in the electropherogram to highlight the data for that peak.

Peak State	Peak Number	Mig. Time (secs)	Mol. Wt. (kDa)	Corrected Area	Area Ratio (to Upper Marker)	Concentration (ng/μl)	Calib. Conc. (ng/μl)	% Total	Observation	Comments
	1	12.26		3.34						
	2	14.66		17.34						
	3	15.70		20.67						
	4	16.85	1.20	1429.55					Lower Marker	
	5	21.64	7.40	476.03					System Peak	
	6	22.04	7.91	246.87					System Peak	
	7	36.30	71.32	1.19	0.0066	0.49		1.27		
	8	39.89	100.36	92.59	0.5131	38.49		98.73		
	9	54.00	260.00	180.45					Upper Marker	
	10	68.06		0.64						

Sample Peaks Found: 2
Total Concentration: 39.0 ng/μl

Fig. 5.1. Results table showing quantitation data for peaks in a sample. Experion software automatically performs % total and relative concentration determinations for each peak identified in every sample.

To exclude a peak from the % total calculation, right-click on the peak (in either the single-well electropherogram view or Results table) and select **Exclude Peak** in the context menu. The new calculations for the remaining peaks appear in the Results table.

This change affects a selected sample. To exclude multiple peaks from multiple samples, use the **Settings** tab and the **Peak Find Settings** options in the **All Wells Settings** dialog (see Section 6.4, Excluding a Peak From Analysis and 6.5, Changing Peak Finding Parameters) and adjust the peak detection criteria.

5.3 Performing Relative Quantitation

Experion software also automatically performs quantitation of all peaks in a sample by a single-point calibration to the 260 kD upper marker and presents these data in the **Concentration (ng/μl)** column of the Results table.

To view these default relative quantitation results, click on the peak of interest in the electropherogram to highlight the data for that peak.

To perform relative quantitation against a different, user-defined internal standard:

- Add the standard to the samples at a single concentration, and
- Change some of the analysis settings in the software

Selection of the user-defined standard in the software may be done after the run is complete. This enables easy comparison of the relative quantitation results derived by using both the upper marker and the user-defined standard.

To perform relative quantitation with a single concentration of a user-defined internal standard:

1. Prepare the protein samples and reagents as outlined in Section 3.5, except add a known concentration of the internal standard (for example, 500 ng/μl) to each sample. For example, replace half the volume of sample with the internal standard (combine 2 μl sample, 2 μl internal standard, and 2 μl sample buffer).

Do not add the internal standard to the Pro260 ladder.

2. Run the Pro260 analysis.
3. After evaluating the run as described in Section 3.10, select **Analysis > Internal Std. and Std Protein Calibration Curve**.
4. In the **Internal Std. and Std. Protein Calibration Curve** window (Figure 5.2), select **Use User Internal Standard** and enter the name, concentration, and molecular weight (MW) of the internal standard. Click **Apply** and then **Close**. The new concentrations appear in the Results table under the **Concentration (ng/μl)** column (Figure 5.3).

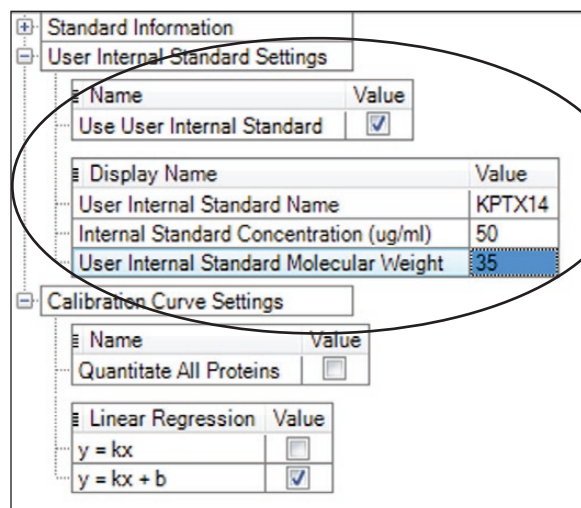


Fig. 5.2. Internal Std. and Std. Protein Calibration Curve window.

Peak State	Peak Number	Mig. Time (secs)	Mol. Wt. (kDa)	Corrected Area	Area Ratio (to Internal Std)	Concentration (ng/ul)	Calib. Conc. (ng/ul)	% Total	Observation	Comments
	1	14.13		58.56						
	2	14.85		32.01						
	3	15.85	1.20	1174.70					Lower Marker	
	4	18.76	5.29	123.32					System Peak	
	5	19.52	6.37	118.01					System Peak	
	6	20.38	7.58	1114.85					System Peak	
	7	26.38	25.42	3.03	0.0021	1.05		3.53		
	8	34.20	73.57	1443.05	1.0000	500.00			User Internal Standard	
	9	37.25	100.00	81.23	0.0563	28.15	45.98	94.75	Calibrated Protein	
	10	38.25	110.01	0.69	0.0005	0.24		0.81		
	11	46.73	209.80	0.79	0.0005	0.27		0.92		
	12	50.50	260.00	78.65					Upper Marker	
	13	63.27		0.66						
Sample Peaks Found 4										
Total Concentration 29.7 ng/ul										
Results Run Summary Protein Search Result Protein Search Criteria Run Info Settings										

Fig. 5.3. Results from relative quantitation.

5.4 Performing Absolute Quantitation

Experion software also allows quantitation using a calibration curve generated by a range of known concentrations of that protein (absolute quantitation). In this method:

- Different concentrations of the protein are separated in different sample wells
- The peak areas of the different concentrations are compared to the peak area of an internal standard. As with relative quantitation, either the upper marker or a user-defined protein can be used as the internal standard
- The protein concentrations in the samples are then determined from this calibration curve

Experion software displays the results for both relative and absolute quantitation, enabling easy comparison of the accuracy achieved by these different approaches.

5.4.1 Absolute Quantitation Using the Upper Marker as Standard

1. Prepare a dilution series of 3–6 different concentrations of the protein used for calibration. The total protein concentration in each sample must be within the linear dynamic range of the assay. Prepare these samples with the other samples for Pro260 analysis as described in Section 3.5.
2. In the **Run Info** tab under **Sample Information**, designate the wells that contain the proteins for the calibration curve by selecting them in the **Use as Standard** column (Figure 5.4). Enter their concentrations in the **Standard Concentration** column and click **Apply**.
3. Run the Pro260 analysis.

Export Import Apply				
Sample Information				
WellId	SampleName	Sample Description	Use as Standard	Standard Concentration (ng/ul)
0	Ladder			
1	Standard 1	A Comment	☒	25
2	Standard 2	A Comment	☒	100
3	Standard 3	A Comment	☒	250
4	Standard 4	A Comment	☒	500
5	Standard 5	A Comment	☒	1000
6	Sample 1	A Comment	☐	0
7	Sample 2	A Comment	☐	0
8	Sample 3	A Comment	☐	0
9	Sample 4	A Comment	☐	0
10	Sample 5	A Comment	☐	0
Study Information System Information				
Results Run Summary Protein Search Result Protein Search Criteria Run Info Settings				

Fig. 5.4. Run Info tab illustrating the creation of a 5-point calibration curve with wells 1–5.

4. After evaluating the run as described in Section 3.10, select **Analysis > Internal Std. and Std Protein Calibration Curve**.
5. In the **Internal Std. and Std. Protein Calibration Curve** window (Figure 5.5), do the following:
 - a. Under **Standard Information**, next to **Standard Protein Molecular Weight**, enter the MW corresponding to the region where the software searches for a peak, for both the calibrants and the samples. The software checks for a peak in the sample wells that has the entered value ($\pm 5\%$). If necessary, modify the concentrations of the calibrants by double-clicking on the concentration value and typing in a new value.
 - b. Ensure that **Use User Internal Standard** is not selected.
 - c. Under **Calibration Curve Settings**, if the protein used for calibration (calibrant) is different or has a different mass than the protein being quantitated, select **Quantitate All Proteins** and click **Apply**. If you do not do this, the software only reports absolute concentrations for peaks that are the same size as the calibrant.
 - d. Under **Calibration Curve Settings**, select the **Linear Regression** to be used.
 - e. Click **Apply** to update the calculation. The calibration curve appears and displays the equation for the linear regression (Figure 5.5). The table below the graphic displays data for the wells containing the standards. Click **Close**.
6. In the **Results** tab, the concentration for each protein identified by Experion software is derived by relative quantitation against the 260 kD upper marker and automatically appears in the **Concentration** column; the calibrated concentrations of proteins found in the specified region appear in the **Calib. Conc.** column.

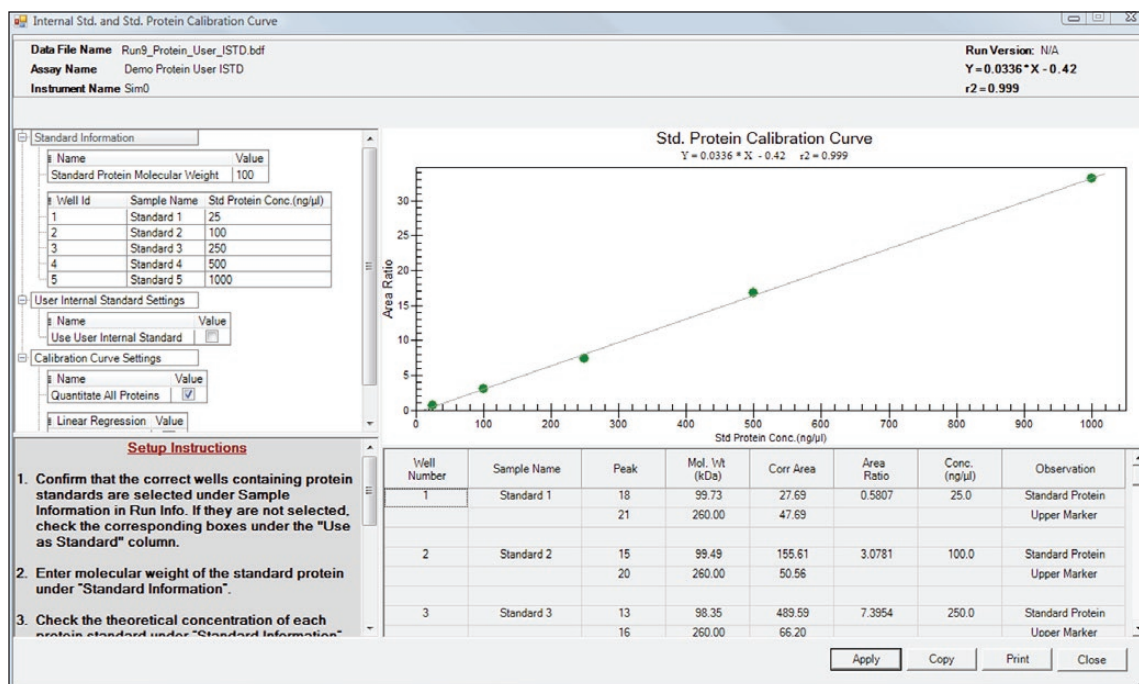


Fig. 5.5. Internal Std. and Std Protein Calibration Curve window. In this example, a 100 kD protein is used to build a 5-point calibration curve for quantitating a 100 kD sample protein.

5.4.2 Absolute Quantitation Using a User-Defined Internal Standard

This is the most stringent method of quantitation. It combines two of the approaches described above:

- A calibration curve is generated using known concentrations of a protein, and
 - A user-defined protein is used as the internal standard
1. Prepare a dilution series of 3–6 concentrations of the protein that you will use for calibration.
 2. Add a known concentration of the internal standard to each sample (except the Pro260 ladder), including those that will be used for calibration. Prepare these samples and other samples as described in Section 3.5.
 3. Under the **Run Info** tab, designate the wells that contain the proteins for the calibration curve and enter their concentrations (Figure 5.5). Click **Apply** and run the Pro260 analysis.
 4. After evaluating the run as described in Section 3.10, select **Analysis > Internal Std. and Std Protein Calibration Curve**.
 5. In the **Internal Std. and Std. Protein Calibration Curve** window (Figure 5.5), do the following:
 - a. Under **Standard Information**, next to **Standard Protein Molecular Weight**, enter the MW corresponding to the region where the software searches for a peak, for both the calibrants (standards) and the samples. The software checks for a peak in the sample wells that has the entered value ($\pm 5\%$). If necessary, modify the standard concentrations by double-clicking on the concentration and entering a new value.
 - b. Select **Use User Internal Standard** and enter the name, concentration, and size of the internal standard. Click **Apply**.
 - c. Under **Calibration Curve Settings**, if the protein used for calibration (calibrant) is different or has a different mass than the protein being quantitated, select **Quantitate All Proteins** and click **Apply**. If you do not do this, the software only reports absolute concentrations for peaks that are the same size as the calibrant.
 - d. Under **Calibration Curve Settings**, select the **Linear Regression** to be used. Click **Apply** to update the calculation.
 6. The calibration curve appears, displaying the equation for the linear regression. The table below the graphic displays the data for the protein standard samples. Click **Close**.
 7. In the Results table, the concentration for each protein derived by relative quantitation against the user-defined internal standard appears in the **Concentration** column, and the calibrated concentration (using the calibration curve in conjunction with the user-defined standard) appears in the **Calib. Conc.** column.



Changing Analysis Settings and Parameters

Once an Experion™ analysis is complete and you have examined the quality of the run and of the data, you can reanalyze the data by changing some of the settings. Most of the steps outlined in this chapter are used for troubleshooting purposes or are optional; they are not required in most analyses.

6.1 Designating and Searching for Specific Proteins

1. Designate and search for specific proteins in your samples:
 - a. In the **Protein Search Criteria** tab, click **Add Protein Name** , or in the electropherogram, right-click on a peak and select **Add to Protein Search Criteria**. The **Protein Search Criteria** tab opens and contains the mass and tolerance range for that entry.
 - b. Click in the field under **Protein Name** and enter the protein name.
 - c. Click in the field under **Mol Wt** and enter the size of the protein in kD.
 - d. Click in the field under the $\pm\sigma$ **mass** column to enter the new value for the tolerance in kD and exclude wells that are not relevant to the search.
2. A list of all the peaks that fit the criteria appears in the **Protein Search Result** tab. Separate tables appear for each protein designated in the search criteria list. To exclude a peak, right-click on the peak name in the table and select **Exclude Peak**.
3. To delete designated proteins ("protein names") from a search, click **Delete Protein Name**  in the toolbar of the **Protein Search Criteria** tab.

6.2 Changing Protein Quantitation Parameters

Experion software uses relative quantitation against the upper marker to calculate the concentrations of all the peaks detected in a sample. The calculated concentrations appear in the Results table, under the **Concentration (ng/μl)** column.

For more information about protein quantitation, including about how to perform relative quantitation against a user-defined standard or to perform absolute quantitation using a calibration curve, refer to Appendix A and the instructions provided in Chapter 5, Protein Quantitation Methods.

6.3 Manually Setting a Marker

To designate a peak as a marker (and change the alignment of a sample), right-click on the peak in either the electropherogram (single-well view) or the Results table and select **Manually Set Lower Marker** or **Manually Set Upper Marker** to define the peak as the new lower or upper marker. Do this if no or incorrect peaks have been selected as the lower or upper marker.

Pink triangles in the virtual gel indicate markers. The markers should be aligned across all lanes.

6.4 Excluding a Peak from Analysis

To exclude one or more peaks from analysis of a single sample, right-click on the peak in either the electropherogram (single-well view) or Results table and select **Exclude Peak** from the context menu. This change affects only a selected sample.

To exclude multiple peaks from multiple samples, open the **Settings** tab and use the **Peak Find Settings** options in the **All Wells Settings** dialog (see Section 6.5, Changing Peak Finding Parameters) to adjust the criteria by which peaks are detected.

Excluded peaks are not included in total protein calculations. Therefore, excluding a peak from analysis affects percentage determination (% total calculations) for the other peaks in a sample.

6.5 Changing Peak Finding Parameters

The parameters entered tell the peak-find algorithm whether a peak is significant.

To edit the parameters used by Experion software for finding peaks in a selected sample/well:

1. Select the sample by clicking on the lane in the virtual gel (or on the sample name in the tree view).
2. Under the **Settings** tab, edit the parameters under **Peak Find Settings**:
 - **Slope Threshold** — represents the variation in fluorescence units over time required to detect a peak (accepted values are 0.2–1,000, with lower values yielding more peaks)
 - **Min. Peak Height** — minimum height required for a peak to be integrated (accepted values are 0.1–1,000). Determine the appropriate value for this parameter by zooming in on a small peak in an electropherogram and reading its fluorescence value on the y-axis
 - **Min. Peak Width** — minimum amount of time that must elapse before a peak is recognized (accepted values are 0.1–10, with lower values yielding more peaks)
3. Click **Apply** to apply the changes and reanalyze the data. Click **Save** in the main screen toolbar to save the new conditions or click **Reset to Default** to recover the default settings.

To modify the peak-find settings for all sample wells:

1. Under the **Settings** tab, click the **All Well Settings** tab or select **Analysis > All Wells Settings**.
2. In the **All Wells Settings** dialog, expand **Peak Find Settings** and select the type of modification you would like to apply (see above for descriptions).
3. Click **OK** to apply changes and **Yes** to overwrite the settings. Click **Save** in the main screen toolbar to save the new conditions. To reset to default settings, click **Reset to Default**.

Do not adjust the Peak Find Settings values so that the markers are eliminated from analysis.

6.6 Changing General Settings

These settings appear in the **Settings** tab, in the **All Wells Settings** dialog under **General Settings** (Figure 6.1). They apply to all samples in a run.

- **Data Frequency** — displays the rate of data collection in Hz
- **Display Start Time** — displays the time Experion software begins displaying peaks
- **Display End Time** — displays the time Experion software stops displaying peaks
- **Upper Marker Concentration** — displays the concentration of the upper marker
- **Use Time Corrected Areas** — corrects for peaks of different sizes that pass the detector at different rates
- **Calibrate All Valid Peaks** — select this option if you are running a calibration curve and want Experion software to calibrate all peaks (this is the same as selecting **Quantitate All Proteins** in the **Internal Std. Protein Calibration Curve** window)
- **Filter Width** — determines the width of the polynomial (in sec) to be applied to the data for filtering (noise reduction). Set this variable to less than twice the width of peaks. Valid values are 0.3–10
- **Polynomial Order** — defines the power series applied to fit the raw data. Valid values are 1–10, with lower values yielding smoother curves

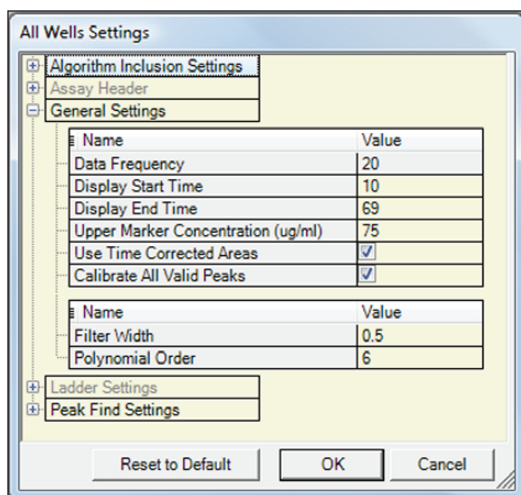


Fig. 6.1. General Settings options.

6.7 Baseline Modification

To change the method used by Experion software for adjusting the baseline in a selected electropherogram (single sample):

- Under the **Settings** tab, select one of the following two options under **Algorithm Inclusion Settings** (Figure 6.2):
 - **Baseline Correction** — (default) subtracts the baseline from the signal. The baseline represents a slow, rolling component in the signal. Generally, this method affects the original shape of the curve and subsequent calculations. The more complex the baseline, the greater impact subtraction might have on the signal. However, for well-resolved peaks with a slow rolling component, this method can normalize calculations across the chip. If peaks are poorly resolved, baseline correction can cause considerable artifacts
 - **Zero Baseline** — subtracts the average baseline value from the signal. This method does not affect the shape of the curve; it shifts the baseline down to zero. If the **Baseline Correction** option results in undesirable artifacts (small peaks do not appear, or too many peaks appear), change this setting to see if it remedies the problem

Deselecting both options makes the original signal(s) appear. Sometimes it is important to see the difference between the signals across the chip if no correction is applied.

- Click **Apply** to apply the changes and reanalyze the data, and click **Save** in the main screen toolbar to save the new conditions or choose **Reset to Default** to recover the default settings.

To edit the parameters used by Experion software for adjusting the baseline for all samples:

- Either open the **Settings** tab or select **Analysis > All Wells Settings**.
- In the **All Well Settings** dialog, expand **Algorithm Inclusion Settings** and select the type of modification you would like to apply (see above for descriptions or Figure 6.2). Click **OK**.
- In the **Experion Application** dialog, click **OK** and **Yes** to overwrite the sample settings. Click **Save** in the main screen toolbar to save the new conditions. To reset to default settings, click **Reset to Default** in the **Settings** tab under **All Well Settings**.

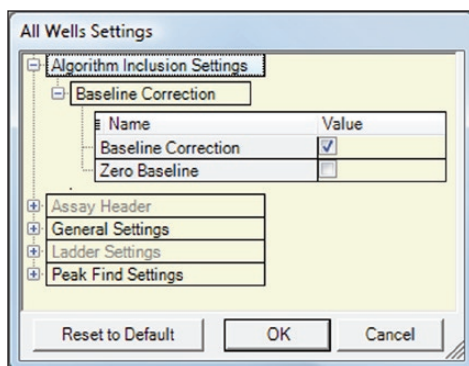


Fig. 6.2. Algorithm Inclusion Settings window.

6.8 Turning Analysis Off

To turn analysis off, select **Analysis > Turn Analysis Off**. Analysis remains off for all runs until it is turned back on.

Turning analysis off displays the raw data from a run; it deactivates alignment, shifting the positions of the upper and lower markers in each sample. In addition, the pink arrows (in the virtual gel) indicating the positions of the markers appear only in the ladder lane. With analysis off, the migration of the markers may not be perfectly aligned and the position of the upper marker may drift (Figure 6.3). Turning analysis off can help reveal if a problem is isolated (one or a few wells have markers out of alignment) or is general (alignment problems across the entire chip).

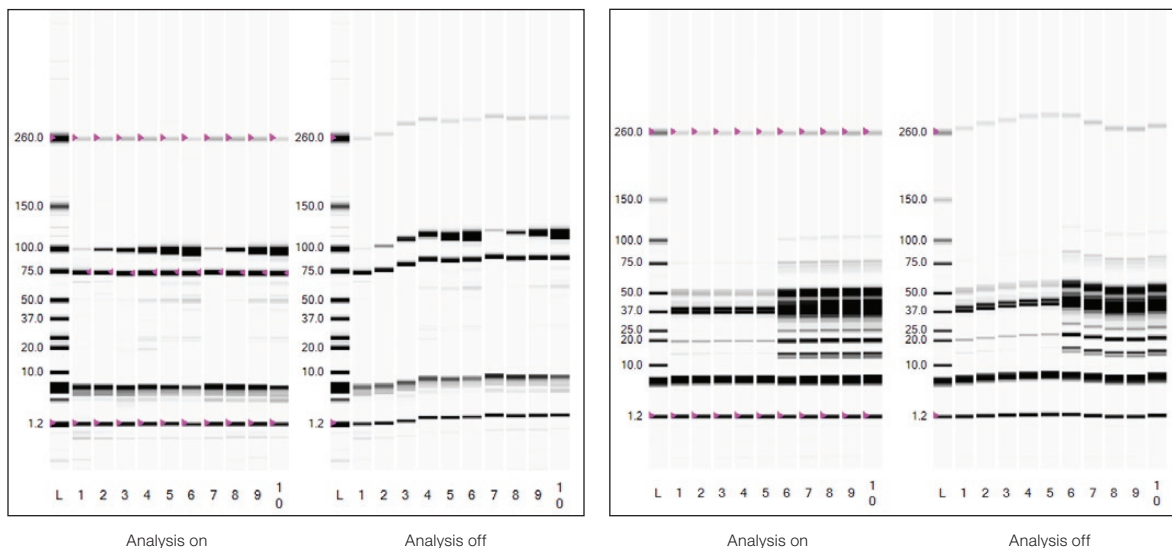


Fig. 6.3. Effects of turning analysis off for two samples.

6.9 Manual Peak Integration

This function enables definition or refinement of peak areas for the ladder and all sample lanes.

1. Select a sample and click  or select **Analysis > Manual Integration** to start manual integration.
2. Adjust the baseline by clicking and dragging the dots at the ends of the peaks (Figure 6.4).
If a peak in the electropherogram is not detected by the software, right-click on that peak and select **Add Peak**. Zoom in on the peak for a better view of the baseline.
3. Save any changes by selecting **File > Save Run As** (use a new file name), or revert to the original baseline and integration by clicking .

Manual integration applies to individual sample wells. Moving back to automated integration loses changes made during manual integration unless those changes are saved.

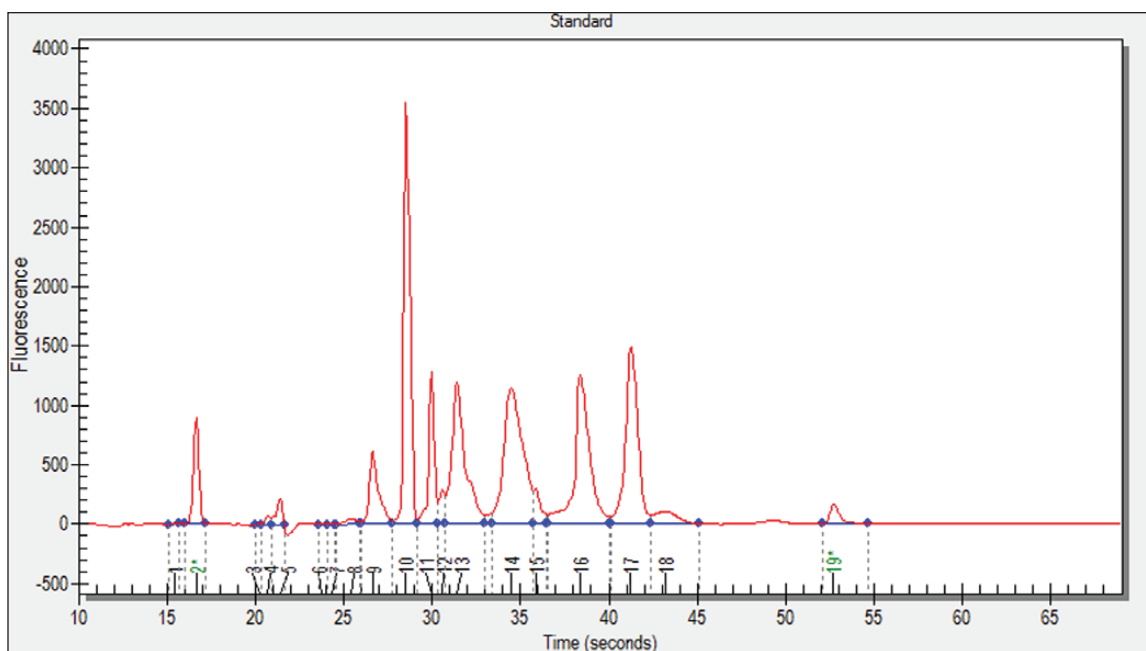


Fig. 6.4. Manual integration mode indicates the beginning and end of peak baselines with blue dots.

7 Troubleshooting

7.1 Electrophoresis and Priming Stations

Error	Probable Cause	Recommended Action
IV Check Failure (Experion™ software)	One or more wells are not filled correctly, and the electrodes are not immersed	Fill all sample wells of the chip with blank or replicate samples Ensure pipets are properly calibrated
	Air bubbles are interfering with the electrical contact in one or more of the wells	Stop the run, remove the chip, and use a clean pipet tip to remove the sample and apply a new aliquot When pipetting, insert the tip vertically and to the bottom of the well. Dispense the liquid slowly. Do not expel air at the end of the pipetting step. Dispense only to the first stop on the pipet. Positive (reverse) pipetting may also be used
	An electrode is loose or disconnected	Perform a conductivity test with the cleaning chip or conductivity chip from the Experion validation kit (catalog #700-7051). Add 850 µl water to the chip, place the chip in the electrophoresis station, and close the lid. In Experion software, select Tools > Diagnose Instrument and select Conductivity in the Instrument Diagnostics dialog If the test passes, focus on the sources of error above. If it fails, contact technical support with the results
	Lid is not closed	Close the lid
	The chip is not primed properly	Repeat the analysis using a new chip
Check Seal (Priming station)	The priming seal is worn, torn, or clogged	Replace the priming seal (catalog #700-7031)
	The chip is not positioned correctly in the priming station	Align the arrow on the chip with the arrow on the platform
	The gel-stain solution (GS) was added to the wrong well for priming	The priming well is highlighted on the chip. See Section 3.6, Prime the Chip
No chip detected	Chip not positioned properly, empty wells, or incorrect reagents	Make sure chip is positioned properly and correct reagents used
Run aborts before all wells are analyzed	The computer power saver options are causing the runs to abort (this is only a problem with Experion software prior to version 3.0)	Upgrade to the newest software version Confirm that all power saver and/or power options are turned off before starting the assay. Do not let the computer “sleep” during a run. If the computer goes to sleep, the run will abort

7.2 Experion Pro260 Analysis

Error	Probable Cause	Recommended Action
Concentrations are not reproducible (CV > 20%)	Sample preparation procedures are inconsistent	Use the same reagents and follow the same procedures for each sample and each run. Do not modify the protocol outlined in Chapter 3 to compensate for protein concentration or sample variability
Peaks were not detected in the Pro260 ladder	An air bubble is at the bottom of the ladder well or microchannels due to poor priming or sample loading; air bubbles can interfere with the contact between the electrode and the liquid	Check the chip for air bubbles after priming and sample loading: - After priming: turn the chip over and make sure the microchannels are filled with gel-stain solution (GS). If the chip is primed correctly, the wells should be hard to see. Compare the primed chip to an unprimed chip if necessary. Do not touch the glass - After sample loading: tilt the chip and view it from different angles against a light background

Error	Probable Cause	Recommended Action
No peaks were detected in some samples	There is not enough sample in the wells	Ensure that pipets are calibrated and that 6 µl solution has been added to the wells
	Samples were not prepared properly	Review the sample preparation procedure detailed in Chapter 3. The Pro260 ladder must be prepared with reducing sample buffer even when samples are to be separated under nonreducing conditions
	Pro260 stain was photobleached	Protect the Pro260 stain, gel-stain solution (GS), sample buffer, and prepared and diluted samples from light Prepare new GS; use a new tube of Pro260 stain if necessary
	Particulates are clogging the microchannels	Use only high-quality, 0.2 µm-filtered water (such as ReadyPrep™ proteomics grade water, not autoclaved water) Verify that the gel (G) and GS were properly filtered If using samples that contain particulates, perform a quick microcentrifuge spin of the prepared sample to pellet particulates before loading
No peaks detected in any samples	Gel-stain solution (GS) is old or has been photobleached	Prepare fresh GS
	Chip was not primed with the correct pressure and time settings	Check settings on the priming station and repeat the analysis with a new chip
	Particulates are clogging the microchannels	Use only high-quality, 0.2 µm-filtered water (such as ReadyPrep proteomics grade water, not autoclaved water) Verify that the gel (G) and gel-stain solution (GS) were properly filtered If using samples, such as cell lysates, that contain particulates, perform a quick microcentrifuge spin of the prepared sample to pellet particulates before loading
Lower and/or upper marker is missing	Samples do not contain sample buffer or were not prepared properly (if both are missing)	Review the sample preparation procedure in Chapter 3
	Late migration	If only the upper marker is missing, refer to the troubleshooting tips for late migration (below)
Electropherogram peaks are much smaller than expected (peaks are <50% of what is expected)	Too much stain in the gel-stain solution (GS); sample cannot destain completely, and the background is high; too much stain can contribute to high background (>1,000 RFU) and low sample and ladder peaks	Prepare new GS using a new tube of stain. Ensure stain is thawed completely before preparation
	Not enough sample was added	Ensure pipets are calibrated Use pipets that accurately deliver volumes of 10 µl or less
	Too much salt in the sample; salt ions compete with the sample ions during injection	Analyze a dilution series of the sample in water on another chip. If the peak heights and calculated protein concentrations of the diluted sample increase, the salt is too high in the original sample. Determine the proper dilution to minimize the salt effect
	One or more components in the sample is beyond the concentration range listed in the compatibility table (Appendix C)	Analyze a dilution series of the sample in water on another chip. Adjust your sample preparation procedure as necessary
	Gel-stain solution (GS) is old or has been photobleached	Prepare fresh GS

Error	Probable Cause	Recommended Action
	Reagents, specifically the Pro260 stain, were not brought to room temperature prior to use; if stain is not equilibrated to room temperature, it will be more concentrated when it is added to the gel, which may result in significantly lower protein peaks and areas Gel (G) and gel-stain solution (GS) were mislabeled or used improperly. The Pro260 chip contains specific wells for G and GS: GS wells carry the sample through the microchannels and supply the dye for analysis, and the G well is used for destaining prior to detection. If G and GS are added to the incorrect wells, peaks can be small, crowded, or missing in electropherograms	Repeat the analysis with a new chip, new tube of stain, and properly equilibrated reagents Ensure chip was primed with GS Ensure G was used in the G well and GS was used in all wells marked GS
Peaks are small, broad, or missing in some samples, but present in other samples on the chip	Pro260 ladder and samples were not completely denatured Contaminants are present	Heat samples and Pro260 ladder for 3–5 min at 95–100°C before loading Check chemical compatibility charts (Appendix C) and prepare samples in a different extraction buffer or dilute them in water, if necessary Do not use autoclaved water for diluting protein samples or ladder. Use only high-quality, 0.2 µm-filtered water (such as ReadyPrep proteomics grade water)
Late migration (peaks broaden and are delayed over the course of the run)	Sample pH is lower than 5 Air bubbles are interfering with electrical contact in one or more of the wells	Turn analysis off (see Section 6.8, Turning Analysis Off, for more information). Late migration is indicated if there is a drift in the migration of the lower marker across the chip and the upper marker migrates later and runs off the top of the gel view. Refer to the troubleshooting tips that follow Adjust pH of the sample to 5 or above Stop the run, remove the chip, and use a clean pipet tip to remove or dislodge the bubbles, or remove and replace the solution in affected wells When pipetting, insert the tip vertically and to the bottom of the well. Dispense liquids slowly. Do not expel air at the end of the pipetting step. Dispense only to the first stop
Peaks migrating faster than usual; electropherogram appears compressed	Electrophoresis station temperature is inappropriate or fluctuating during the run (should be 30–35°C); there is no cooling unit in the electrophoresis station, so if the temperature changes during the course of a run, the samples will exhibit different separation characteristics	Ensure that the temperature of the room is appropriate and stable, and place the electrophoresis station away from all heat sources, such as windows or ovens
Molecular mass (sizing) is incorrect	Pro260 ladder peaks and upper and lower markers were improperly assigned by the software	Exclude peaks (follow the instructions in Section 6.4, Excluding a Peak From Analysis) or manually assign markers (follow the instructions in Section 6.3, Manually Setting a Marker), if necessary Sizing should be within 10% of the size seen by SDS-PAGE, except for proteins with high levels of posttranslational modification, or highly acidic or basic proteins

Error	Probable Cause	Recommended Action
Protein quantitation is incorrect	Upper and lower markers were not properly assigned by the software Pipetting and/or dilution errors occurred during sample preparation or chip loading Samples are old or have not been stored properly, and the proteins degraded Samples were not denatured properly; regardless of whether reducing or nonreducing conditions are used, the Pro260 ladder and samples must be heat-denatured for the assay to run properly Samples do not have the same staining properties as the upper marker	Check that both markers were properly assigned. Manually select the marker(s), if necessary (follow the instructions in Section 6.3, Manually Setting a Marker) Ensure your calculations and dilutions are correct Ensure that pipets are calibrated, and use pipets that accurately deliver volumes of 10 µl or less Do not modify the sample preparation and loading protocols described in this manual Ensure that the total protein concentration of your samples (including any user-defined internal standards that may be added) is within the linear dynamic range before proceeding with the sample preparation instructions described in Section 3.4; if total protein concentration is unknown, run a Pro260 analysis of a dilution series of your sample Review the essential practices described in Chapter 2 before initiating another analysis with a new chip Prepare fresh samples and try the analysis again Heat the samples and the Pro260 ladder for 3–5 min at 95–100°C before loading them into the chip. Keep the caps of the sample and ladder tubes closed during heating to prevent loss of vapor and subsequent changes in concentration. Briefly centrifuge the samples before opening Use a different internal standard or a calibration curve (see Chapter 5, Protein Quantitation Methods, for more information)
Ghost peaks or contaminants appear in electropherograms	Electrodes are contaminated	Follow the deep cleaning procedure described in Appendix B and the software Help section (search term “electrodes”) Use only 0.2 µm-filtered water (such as ReadyPrep proteomics grade water) for diluting the protein samples and Pro260 ladder; do not use autoclaved water
Upper or lower marker is incorrectly identified	Sometimes Experion software selects the incorrect peak to correspond to the upper or lower marker. Such an error affects alignment and sizing	Manually select the upper and lower markers by following the instructions in Section 6.3
Portion of the electropherogram is incorrectly identified as a peak	Default integration settings are inappropriate for the sample	Manually exclude a peak (see Section 6.4) or change the peak finding parameters (see Section 6.5)
Virtual gel does not resemble an SDS-PAGE separation of the same samples	The gel view is a direct interpretation of the electropherogram data. By default, the fluorescence intensity of the bands in each lane is scaled to the intensity/peak height of the highest peak in that sample (scale to local). Consequently, each gel lane may have a different scale of fluorescence intensity. In addition, Experion Pro260 analysis differs from SDS-PAGE in several ways that may impact separations	Adjust scaling of the virtual gel to Scale To Global (see Section 4.2). This also adjusts the scale on all the electropherograms. Adjust the intensity bar to the right of the virtual gel as needed In an Experion Pro260 analysis, the separation of the 10 kD and 260 kD markers is almost linear, whereas in SDS-PAGE it is logarithmic. To mimic the appearance of an SDS-PAGE separation, select Tools > Options > Advanced and select the Gel Mobility Correction option
Glycoproteins are not separating as expected	Large and complex glycosylated proteins may not migrate as expected with the Experion Pro260 assay; however, their separation characteristics will be reproducible	For more information, refer to Bio-Rad bulletin 5453 or contact Technical Support

Error	Probable Cause	Recommended Action
Baseline is fluctuating (for example into very negative and positive values)	Air bubbles are interfering with electrical contact in one or more of the wells	Stop the run, remove the chip, and remove and replace solution in affected wells
	Chip is not completely primed	Prime and load a new chip
	Samples are too concentrated; protein levels are out of the range of the assay	Dilute the samples and repeat the analysis

7.3 Contacting Technical Support

If you are having problems with a run and would like the Bio-Rad Technical Support team to review your data, please submit the following files to **LSG_TechServ_US@bio-rad.com** for analysis:

- Run file (C:\Program Files\Bio-Rad Laboratories\Experion Software\Data\<daily folder name>\<run name>.bdf)
- Run log (C:\Program Files\Bio-Rad Laboratories\Experion Software\Data\<daily folder name>\<run name>.log)
- Packet file (C:\Program Files\Bio-Rad Laboratories\Experion Software\Data\<daily folder name>\Packet\<run name>.txt)
- System log (C:\Program Files\Bio-Rad Laboratories\Experion Software\Backup\Version X\Experion_System.log or .txt)
- Lot numbers of the Pro260 chips and the Experion Pro260 analysis kit

To export the run file, log file, and packet file, select the run(s) in the tree view and click **Export Selected Runs**. The software exports the files into folders with the corresponding project name(s).



Appendices

Appendix A: How the Experion™ System Works

The Experion automated electrophoresis system performs electrophoresis of samples within a microfluidic chip. Within each chip, a series of microchannels connects the sample wells to a separation channel and buffer wells. A set of electrodes in the electrophoresis station applies a voltage across the microchannels, causing charged molecules in the samples to migrate into and through the separation channel. Samples are run sequentially, with a sufficient lag between them to prevent cross-contamination. For separation, the microchannels are filled with a proprietary gel-stain solution (GS) that acts as a sieving matrix, and under denaturing conditions in the presence of lithium dodecyl sulfate (LDS), the sample proteins migrate through the separation channel at a rate based on their size. Finally, proteins interact with a fluorescent dye during separation and are detected as they pass a laser and photodiode detector (laser-induced fluorescence).

Protein analysis is accomplished with the Experion Pro260 analysis kit and involves the following steps:

- **Preparing the chip (priming and loading)** — Priming fills the microchannels of the microfluidic chip with GS, which contains both the sieving matrix and fluorescent dye. Protein samples are then prepared in Pro260 sample buffer, which contains LDS, heat-denatured, and added to the sample wells
- **Running the chip** — The chip is inserted into the electrophoresis station, and as the instrument lid is closed, electrodes come into contact with the solution in the wells. Voltage is applied to the sample wells of the chip, causing the charged, LDS-coated protein ions to migrate into the separation channel. In the separation channel, the different proteins separate as they move at different rates through the gel matrix, depending on their size. During separation, the fluorescent dye associates with the LDS micelles coating the proteins and with free micelles¹
- **Detecting the fragments** — As the molecules migrate toward the end of the separation channel, destaining occurs. Pro260 gel (G), which does not contain dye or LDS, flows alongside the separation channel. Diffusion of free LDS into this detergent-free zone reduces the concentration of LDS below its critical micellar concentration, releasing dye molecules from unbound micelles. When it is free of the hydrophobic interior of the LDS micelle, the dye fluoresces weakly; the background signal is thus reduced. Dye-micelle complexes are more stable to the destaining process when they are bound to proteins. Downstream a laser excites the dye, causing it to fluoresce if it is bound to the LDS micelle-protein complexes. A photodiode detects the fluorescence, and Experion software plots the fluorescence intensity vs. time to produce an electropherogram and a virtual gel image
- **Analyzing the data** — Following separation, Experion software subtracts background noise, removes spikes, identifies and integrates peaks, and assigns their sizes and concentrations. Following analysis, parameters may be changed and the data reanalyzed

¹ An important difference between the Experion system and traditional sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) lies in how sample fragments are detected: in SDS-PAGE, samples are generally stained in the gel once separation is completed, while in the Experion system proteins are stained with a fluorescent dye during separation.

How Experion Software Analyzes Proteins

Data Presentation

As the photodiode detects fluorescent signals from the dye-LDS micelle-protein complexes, Experion software converts the signal into an electropherogram (a plot of fluorescence vs. time). Experion Pro260 electropherograms generally have the following features (Figure A.1):

- Sample peak(s) — signal(s) generated by the sample protein(s)
- System peaks — cluster of signals generated by small molecules that interact with LDS micelles. The system peaks are not considered in the concentration determination
- Upper and lower markers — signals generated by the internal upper (260 kD) and lower (1.2 kD) markers, which are included in the sample buffer to normalize the separation of proteins across all wells in a chip. The upper marker is also used as an internal standard for relative quantitation (see below). For each lane, comparison of the area of this peak with the area of every detected peak allows estimation of relative concentration

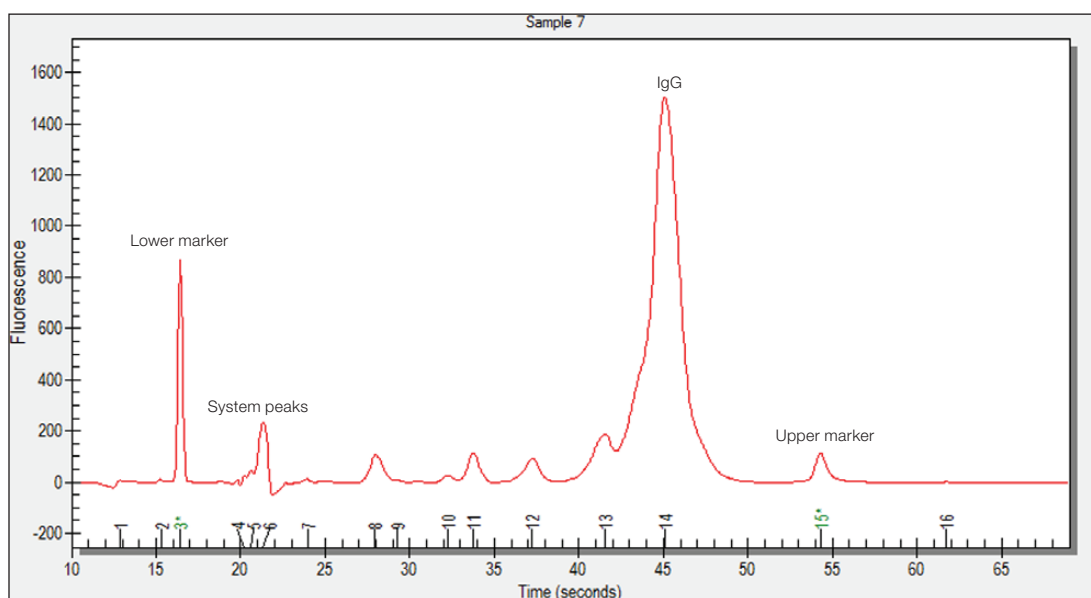


Fig. A.1. Electropherogram generated by Experion Pro260 analysis of immunoglobulin (IgG) under nonreducing conditions. The relative positions of the lower marker, system peaks, sample peak (IgG), and upper marker are shown.

Once separation occurs, the data are converted into a densitometric gel-like image, or virtual gel (Figure A.2). Each lane in the virtual gel corresponds to a different sample. The sample and system peaks and upper and lower markers seen in the electropherogram also appear in the virtual gel.

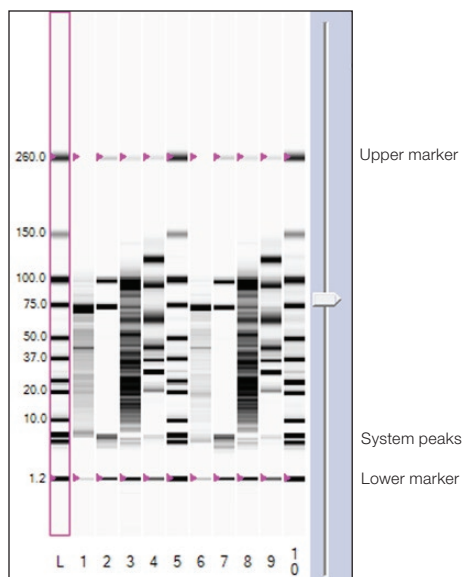


Fig. A.2. Virtual gel generated by Experion Pro260 analysis. Shown are separations of the Pro260 ladder (L) and 10 samples on a single chip.

Once separation is complete, the software subtracts background noise, removes spikes, identifies and integrates peaks, and assigns their sizes and concentrations. The results of data analysis are tabulated and presented in a Results table at the end of analysis (Figure A.3). The types of data available include the following:

- Protein size (or mass, in kD)
- Protein concentration (in ng/μl): relative concentration under **Concentration (ng/μl)** and absolute concentration under **Calib. Conc. (ng/μl)**
- % Total sample (percentage determination)

Peak State	Peak Number	Mig. Time (secs)	Mol. Wt. (kDa)	Corrected Area	Area Ratio (to Upper Marker)	Concentration (ng/μl)	Calib. Conc. (ng/μl)	% Total	Observation	Comments
	1	14.63		38.75						
	2	15.69		12.15						
	3	16.85	1.20	1089.70					Lower Marker	
	4	21.70	7.47	445.92					System Peak	
	5	26.34	18.28	1.43	0.0129	0.97		8.14		
	6	32.60	47.34	0.90	0.0082	0.61		5.16		
	7	36.23	70.88	1.17	0.0106	0.80		6.70		
	8	39.87	100.16	12.13	0.1100	8.25	48.95	69.24	Standard Protein	
	9	40.83	109.12	1.89	0.0171	1.28		10.77		
	10	54.00	260.00	110.30					Upper Marker	
Sample Peaks Found 5										
Total Concentration 11.9 ng/μl										
Results Run Summary Protein Search Result Protein Search Criteria Run Info Settings										

Fig. A.3. Results table generated by Experion Pro260 analysis.

Data Analysis: Normalization, Sizing, and Quantitation

Following separation, Experion software analyses proteins using one or more of the following:

- Internal markers for normalizing the migration times of samples in the different wells
- Pro260 ladder for determining fragment size (sizing)
- Internal markers or calibrants for determining concentration (quantitation)

Normalization: Aligning the Protein Peak Migration Times

To compensate for small variations in each separation, Experion Pro260 analysis uses internal markers to normalize the migration times among samples. The two internal markers, an upper marker (260 kD) and lower marker (1.2 kD), are included in the Experion Pro260 sample buffer. Therefore, both of these markers are added to each sample and the Pro260 ladder (Figure A.1). Inclusion of these markers and the normalization process ensure that the software accurately identifies and sizes peaks.

Protein Sizing: Determining Molecular Mass

The first sample analyzed is the Pro260 ladder, a modified version of the Precision Plus Protein™ standards that has been optimized for automated electrophoresis on the Experion system. The Pro260 ladder contains nine purified recombinant proteins of 10–260 kD. Experion software constructs a standard curve of migration time as a function of size from the ladder separation. It then calculates the size of the proteins from the sample wells by comparing their migration times to the curve.

Protein Quantitation: Determining Protein Amount

Experion software offers two different types of protein quantitation methods: percentage determination and concentration determination. All measurements are based on the time-corrected peak area (corrected area) of each peak identified in an electropherogram. The corrected area of a peak is proportional to the amount of the protein it represents in a mixture. For more details of the protein quantitation methods used by Experion software, refer to bulletin 5784.

Percentage determination — measures the percentage of each protein in a mixture (% total). This method is commonly used for determining protein content, protein purity, and protein stability. The accuracy of this method depends on the dye binding efficiency of each component in the protein mixture. As with other dye-based assays or other methods of quantitation using SDS-PAGE, differences in the amino acid sequences or structures of proteins result in their unique interaction with the Pro260 stain-LDS micelle complexes, which in turn affects peak intensity. As a result, the % total value may not always reflect the actual percentage in mass for each protein in a mixture. If the component is well-characterized, this method of quantitation can be an efficient method for processes requiring routine monitoring of protein samples, such as protein manufacturing and purification

Concentration determination — provides the amount of the protein(s) in a mixture rather than just a percentage of the total. There are several different ways that concentration determination can be performed, and these methods provide more or less accuracy depending on the type of internal standard used and the extent to which a calibration curve is used:

- **Relative quantitation** — Experion software uses an internal standard to estimate the concentrations of sample proteins. Estimates are calculated by comparing the corrected peak area of the sample peak to that of the internal standard in each sample well. The internal standard can be either the upper marker (260 kD) or a user-defined internal standard (between 10 and 260 kD) added to each sample at a known concentration
- **Absolute quantitation** — Experion software can also create a calibration curve to use for quantitation. To create the calibration curve, a protein calibrant at different concentrations is separated in different sample wells. The corrected areas of the different sample peaks are compared to that of an internal standard (as with relative quantitation, the upper marker or a user-defined protein can be used). Experion software plots the resulting ratios as a function

of protein concentration, and the sample protein concentration is then determined from this calibration curve and reported in the the **Calib. Conc.** column of the **Results** tab. Absolute quantitation methods are often more accurate than relative concentration methods

How Experion Pro260 Analysis Differs from SDS-PAGE

Though the Experion Pro260 analysis kit and SDS-PAGE both use a gel-based matrix and voltage difference to separate proteins according their size, there are a number of fundamental differences between the two methods that may affect how specific proteins are separated and how data are interpreted (see Table A1 below or refer to bulletins 5299, 5423, and 5453).

Table A.1. Comparison of Experion Pro260 analysis and SDS-PAGE.

Aspect	SDS-PAGE	Experion Pro260 Analysis
Sieving polymer (matrix)	Formulation can be changed to affect resolution	Fixed formulation
	Crosslinked polymer	Polymer (not crosslinked)
Protein stain	Multiple available	Single, proprietary stain
	Binds proteins	Binds LDS micelle-protein complexes
Detection	Separate staining step, proteins stained after separation	Integrated separation step; proteins stained during separation
	Separate destaining step	Integrated destaining; proteins destained after separation
	Requires photography or other imaging equipment for documentation, analysis	Laser excites the stain, detection using a photodiode, both incorporated
Types of separation	Denaturing and native; reducing and nonreducing conditions	Denaturing; reducing and nonreducing conditions
Downstream applications	Separated proteins available for excision, electroelution, blotting applications	Proteins not available for further analysis
Reagent and sample usage	Moderate to extensive (ml to L)	Minimal (µl)
Waste generation	Moderate to extensive (ml to L)	Minimal (µl)
Overall analysis time	Hours (depends on gel size, staining method, and methods of documentation and analysis)	30 min/chip (10 samples per chip)
Data analysis	Manual; requires additional equipment and software	Automated (with flexibility of manual adjustments)

Appendix B: Deep Cleaning Procedure

This manual procedure applies Experion™ electrode cleaner directly to the 16 platinum pins in the electrode manifold. It is the most stringent method of cleaning the Experion electrode pins and helps ensure RNase-free operation of the system.

The need for this deep cleaning will vary from lab to lab and depends on the frequency of use of the Experion system and on the types of assays (protein, RNA, or DNA analysis) that are performed.

Perform deep cleaning:

- Prior to first use of the Experion electrophoresis station
- Between a protein and DNA or RNA analysis (an Experion electrophoresis station used to analyze protein samples will become contaminated with RNases and should be cleaned prior to any RNA analysis), or between a DNA and RNA analysis
- Whenever RNase contamination is suspected or other contamination (for example, salt deposits or other precipitates) is visible on the electrodes
- Whenever a chip has been left in the electrophoresis station for an extended period of time (for example, overnight)

Perform this procedure with the electrode manifold in place (installed in the electrophoresis station).

It requires the following supplies:

- Experion electrode cleaner (catalog #700-7252)
- Experion DEPC-treated water (catalog #700-7253)
- Foam cleaning swabs (catalog #700-7264)

Warning: Keep the chip platform dry during this procedure. Either cover the chip platform using plastic wrap and a paper towel or foil or place a cleaning chip on the chip platform.

1. Shut off power to the electrophoresis station.
2. Add 0.5–1 ml Experion electrode cleaner to a microcentrifuge tube. Insert a swab into the solution until it is saturated with the electrode cleaner.
3. Use the swab to gently scrub each electrode pin one at a time, on all four sides. Do not press too hard on the pins, as they may bend. Finally, clean the tip of each pin. Add more electrode cleaner to the swab as necessary.

-Or-

Move the swab up and down, and side to side, 2–3 times along the columns and rows of pins. Finally, clean the tip of each of pin. Add more electrode cleaner to the swab as necessary.

4. Repeat steps 2 and 3 twice, each time with a fresh swab soaked in DEPC-treated water.
5. Fill a cleaning chip with 800 µl DEPC-treated water, place it in the electrophoresis station, and close the lid for 5 min.
6. Remove the chip and allow the pins to dry completely by leaving the lid open for 5–10 min.

Refer to the Experion system manual for more details and images.

Appendix C: Chemical Compatibility

For best results, the starting protein sample (undiluted) should contain components at concentrations less than or equal to the values indicated in the tables below. If concentrations are above these values, dilute the samples accordingly.

Buffers

Table C.1. Buffers with no observable effect on Experion™ Pro260 separation or analysis at or below the concentrations listed.

Buffer	Concentration
B-PER (Pierce)	0.5x
BugBuster (Novagen)	1x: BugBuster and benzonase nuclease
CPEB (ReadyPrep™ protein extraction kit, cytoplasmic/nuclear)	0.5x
GST binding/wash buffer (Novagen)	0.5x
His elution buffer (Pierce)	0.5x
His wash buffer 1 (Pierce)	0.5x
His wash buffer 2 (Pierce)	0.5x
IEF cathode buffer	1.5x: 30 mM Lysine, 30 mM arginine
IMAC elution buffer	1x: 50 mM KH ₂ PO ₄ , 150 mM NaCl, 250 mM imidazole, pH 8.0
IMAC lysis buffer	1x: 25 mM KH ₂ PO ₄ , 150 mM NaCl, pH 8.0
IMAC wash buffer 1	1x: 100 mM KH ₂ PO ₄ , 600 mM KCl, 10 mM imidazole, pH 8.0
IMAC wash buffer 2	1x: 100 mM KH ₂ PO ₄ , 600 mM KCl, 20 mM imidazole, pH 8.0
LB medium	0.5x: 1% Tryptone, 0.5% yeast extract, 1% NaCl
M1 (ReadyPrep protein extraction kit, membrane I)	0.5x
M2 (ReadyPrep protein extraction kit, membrane I)	0.5x
M-PER	0.5x
PBS	1.5x: 15 mM Sodium phosphate, 225 mM NaCl, pH 7.5
PBS-Tween	1x: 10 mM Sodium phosphate, 150 mM NaCl, 0.05% Tween, pH 7.2
S1 (ReadyPrep protein extraction kit, signal)	0.5x
Tricine sample buffer	0.5x: 100 mM Tris-HCl, pH 6.8, 1% SDS, 20% glycerol, 0.02% Coomassie Brilliant Blue
Tris-borate-EDTA (TBE)	1.5x: 133 mM Tris, 133 mM boric acid, 3 mM EDTA, pH 8.3
Tris/CAPS	1.5x: 90 mM Tris, 60 mM CAPS, 0.15% SDS pH 9.6
Tris/glycine	1.5x: 37.5 mM Tris, 288 mM glycine, pH 8.3
Tris/glycine/SDS	1.5x: 37.5 mM Tris, 288 mM glycine, 0.15% SDS, pH 8.3
Tris/Tricine/SDS	1.5x: 150 mM Tris, 150 mM Tricine, 0.15% SDS, pH 8.3
Urea/phosphate/NaCl buffer	0.5x: 4 M Urea, 25 mM sodium phosphate, 150 mM NaCl, pH 8
Criterion™ XT MOPS buffer	1.5x
Criterion XT sample buffer	1.5x
Criterion XT Tricine buffer	1.5x
Zymogram sample buffer	0.5x: 31.3 mM Tris-HCl, pH 6.8, 2% SDS, 12.5% glycerol, 0.005% bromophenol blue

Table C.2. Buffers with mild effects on Experion Pro260 separation or analysis.

Buffer	Concentration	Effect(s)
IEF anode buffer	1.5x: 10.5 mM phosphoric acid	Quantitation
Laemmli buffer	0.5x: 31.3 mM Tris-HCl, pH 6.8, 1% SDS, 12.5% glycerol, 0.005% bromophenol blue, 2.5% β -mercaptoethanol	Quantitation
PSB (ReadyPrep protein extraction kit, membrane I)	0.5x	Quantitation
RIPA buffer	1x: 150 mM NaCl, 1.0% NP40, 0.5% sodium deoxycholate, 0.1% SDS, 50 mM Tris	Enlarged system peak shifts to right; 10 kD sizing affected
TBS	1.5x: 30 mM Tris, 750 mM NaCl pH 7.4	Quantitation

Table C.3. Buffers with severe effects on Experion Pro260 separation or analysis.

Buffer	Effect(s)
Rehydration buffer (ReadyPrep 2-D starter kit)	Enlarged system peak shifts to right; 10 kD peak is missing, sizing affected
S2 (ReadyPrep protein extraction kit, signal)	Enlarged system peak shifts to right; 10 kD peak is missing, sizing affected
Reagent 2 (ReadyPrep sequential extraction kit)	Enlarged system peak shifts to right; 10 kD peak is missing, sizing affected
Reagent 3 (ReadyPrep sequential extraction kit)	Enlarged system peak shifts to right; 10 kD peak is missing, sizing affected
Total protein extraction buffer	Enlarged system peak shifts to right; 10 kD peak is missing, sizing affected
Criterion XT MES buffer (1.5x)	10 kD peak is missing

Chemicals

Table C.4. Chemicals with no observable effects on Experion Pro260 separation or analysis at or below the concentrations listed.

Chemical	Concentration	Chemical	Concentration
Acetonitrile	30%	MeOH	15%
Ammonium sulfate	250 mM	Na_3VO_4	5 mM
Arginine	500 mM	NaCl	250 mM
DMSO	15%	NaF	50 mM
EDTA	10 mM	NaHCO_3	50 mM, pH 8
EGTA	150 mM	NaN_3	0.75%
Ethanol	30%	NaOAc	50 mM, pH 4
Glucose	0.5 M	NaOH	150 mM
Glycerol	35%	NDSB201	1%
Glycine	1 M	Sodium carbonate	150 mM, pH 11.5
Guanidine isothiocyanate	25 mM	Succinic acid	100 mM
HEPES	50 mM, pH 7	Tetramethylammonium chloride	1 M
Imidazole	0.5 M, pH 7	Trehalose	5%
KCl	250 mM	Tris base	1.5 M
MgCl_2	150 mM	Urea	4.5 M

Table C.5. Chemicals with mild effects on Experion Pro260 separation or analysis.

Chemical	Concentration	Effect(s)
Acetone	15%	Quantitation affected
Bio-Lyte® 3/10	0.75%	Quantitation affected
Guanidine HCl	25 mM	Enlarged system peak shifts to right; 10 kD sizing affected
KCl	1.5 M	Quantitation affected
MES	50 mM, pH 6.0	Quantitation affected
MOPS	100 mM, pH 7.2	Quantitation affected
PEG	0.50%	Quantitation affected
Propylene glycol	50%	Enlarged system peak shifts to right; 10 kD sizing affected
Sucrose	200 mM	Quantitation affected

Table C.6. Chemicals with severe effects on Experion Pro260 separation or analysis.

Chemical	Concentration	Effect(s)
Ammonium sulfate	1.5 M	Protein precipitation
Guanidine isothiocyanate	1 M	Protein precipitation; no markers appear
Guanidine HCl	1 M	Protein precipitation; no markers appear
HCl	150 mM	Protein precipitation; markers missing
Pluronic F-68	0.1%	Protein smears

Detergents and Reductants

Table C.7. Detergents and reductants with no observable effects on Experion Pro260 separation or analysis.

Detergent/Reductant	Concentration
ASB-14	0.10%
CHAPS	0.50%
CHAPSO	0.10%
Deoxycholate	1.0%
Octyl-thioglucopyranoside	0.50%
Tributylphosphine (TBP)	7.5 mM
Triton X-100	0.40%
Tween-20	0.50%

Table C.8. Detergents and reductants with mild effects on Experion Pro260 separation or analysis.

Detergent/Reductant	Concentration	Effect(s)
ASB-14	0.5%	Enlarged system peak shifts to right; 10 kD sizing affected
CHAPS	1%	Enlarged system peak shifts to right; 10 kD sizing affected
CHAPSO	0.2%	Enlarged system peak shifts to right; 10 kD sizing affected
Deoxycholate	2%	Enlarged system peak shifts to right; 10 kD sizing affected
LDS	3%	Enlarged system peak shifts to right; 10 kD sizing affected
NP40	0.5%	Enlarged system peak shifts to right; 10 kD sizing affected
SB3-10	0.1%	Enlarged system peak shifts to right; 10 kD sizing and quantitation affected
SDS	1%	Quantitation affected
Tween-20	0.75%	Enlarged system peak shifts to right; 10 kD sizing affected
Zwittergent 3-14	0.10%	Enlarged system peak shifts to right; 10 kD sizing affected
β-Mercaptoethanol	1.5 M	Quantitation affected
DTT	150 mM	Quantitation affected

Table C.9. Detergents and reductants with severe effects on Experion Pro260 separation or analysis.

Detergent/Reductant	Concentration	Effect(s)
ASB-14	2%	Enlarged system peak shifts to right; 10 kD peak is missing
n-Dodecyl- β -D-maltoside	1%	Enlarged system peak shifts to right; 10 kD peak is missing
NP40	2%	Enlarged system peak shifts to right; 10 kD peak is missing
SB3-10	2%	Enlarged system peak shifts to right; 10 kD sizing affected
Triton X-100	1%	Enlarged system peak shifts to right; 10 kD peak is missing
Tween-20	1%	Enlarged system peak shifts to right; 10 kD peak is missing
Tween-80	1%	Enlarged system peak shifts to right; 10 kD peak is missing
Zwittergent 3-14	1%	Enlarged system peak shifts to right; 10 kD peak is missing

Appendix D: Glossary

% Total

Calculation performed by Experion™ software to show the percent of the total protein comprised by a particular peak; used in purity assessments

Absolute quantitation

Method of quantitation of a defined protein in which concentrations are determined against a calibration curve, which is usually generated using the same protein

All-wells view

Mode of display within the electropherogram view wherein electropherograms of all wells are shown

Baseline

Line of initial data that serves as the control, or signal representing no fluorescence

Calibration curve

Curve of fluorescence intensity (signal) as a function of protein concentration; generated using various known concentrations of a specific protein and used for calculating the protein concentration of samples

Chip

Microfluidic device used to perform separation using the Experion system

Electropherogram

Trace recording of signal (fluorescence intensity) as a function of time; displayed in the electropherogram view

Electropherogram view

Portion of the Experion software user interface dedicated to display of electropherogram data

Electrophoresis station

Instrument that houses all the electrical, optical, and hardware components necessary to perform separation, staining, destaining, band detection, and imaging of samples; equipped with a universal power supply to operate at 100–240 V, 50–60 Hz

Gel priming well

Well of the microfluidics chip used to fill the microchannels of the chip with gel-stain solution (GS) during priming

Gel-stain solution (GS)

A mixture of gel matrix and stain; used for sample separation and detection

Gel view

Portion of the Experion software user interface dedicated to display of virtual gel data

Internal standard

Protein added to the sample for quantitation; can be either the upper marker (260 kD), which is included in the Pro260 sample buffer, or a user-defined protein (that is baseline resolved from the sample peak being quantitated) added to each sample at a known concentration

LabChip

Microfluidic “lab-on-a-chip” technology developed by Caliper Life Sciences; the Experion automated electrophoresis system was built on LabChip technology

LDS micelle-protein complex

Micelle of the detergent lithium dodecyl sulfate (LDS) bound with protein(s)

Lithium dodecyl sulfate (LDS)

Anionic detergent that may be used in place of sodium dodecyl sulfate (SDS) for denaturing electrophoresis

Lower marker

1.2 kD marker included in the Experion Pro260 sample buffer and used to normalize the separation of proteins across all wells in the chips

Microchannels

Fine channels approximately the diameter of a human hair are used for samples separation in Experion microfluidic chips

Microfluidic technology (microfluidics)

Multidisciplinary field comprising physics, chemistry, engineering, and biotechnology that deals with the behavior, precise control, and manipulation of microliter and nanoliter volumes of fluids

Nonreducing

Sample condition that does not include reducing agents for the reduction of disulfide bonds

Normalization

Alignment of protein peak migration times across different sample wells of the same chip; the lower and upper markers are used by Experion software for this purpose

Overlay

Function of Experion software that allows different electropherograms to be superimposed for easier comparison

Peak

Signal in an electropherogram generated and identified by the peak-find settings to be significantly different from the baseline

Priming

Filling the microchannels of the microfluidic chip with gel-stain solution in preparation for analysis

Priming station

Instrument used to prepare chips for sample analysis; applies pressure to the priming well of the microfluidic chip and fills the network of microchannels with the gel-stain solution

Pro260 ladder

Modified version of the Precision Plus Protein™ standards that has been optimized for automated electrophoresis on the Experion system; contains 9 purified recombinant proteins of 10–260 kD

Project folder

User-defined folder containing multiple run files comprising a particular project

Project tree

Hierarchical tree of all saved projects and the data files they contain

Quantitation

Determination of the concentration of a protein in a sample

Reducing

Sample condition that includes reducing agents for the reduction of disulfide bonds

Relative quantitation

Method of quantitation in which concentrations are determined against a single-point internal standard

Results table

Tab that displays analysis results, settings, and parameters

Results and settings

Portion of the Experion software user interface dedicated to display of tabulated analysis results and analysis settings and parameters

Run file

File containing data for a single run

Single-well view

Function of Experion software that allows an electropherogram from a single well to be viewed

Sizing

Determination of the molecular mass of a protein

Spin filter

Polypropylene spin tubes with filter membranes used to remove particulate matter from the gel-stain (GS) and gel (G) by centrifugation

System peaks

Cluster of signals generated by small molecules that interact with fluorescent detergent micelles; not considered in the concentration estimation

Tree view

Portion of the Experion software user interface dedicated to display of the project tree; used to locate and manipulate data files and organize runs into projects

Upper marker

260 kD protein included in the Experion Pro260 sample buffer and used to normalize the separation of proteins across all wells in the chips; also used as an internal standard for relative quantitation by default

User-defined standard

Protein (10–260 kD) added to each sample at a known concentration and used for quantitation

Virtual gel

Densitometric, gel-like image generated from electropherogram data and displaying peaks as bands; displayed in the gel view

Vortex station

Instrument used to mix RNA or DNA samples and loading buffers (supplied in the kits) in the chip wells; used for nucleic acid applications only

Appendix E: Bibliography

Visit us on the Web at www.bio-rad.com (search term “Experion”) for a list of technical literature and to download copies of all current documents.

Bulletin	Title
3176	Monitoring the Expression, Purification, and Processing of GST-Tagged Proteins Using the Experion™ Automated Electrophoresis System
5283	Bio-Rad Applies Microfluidics to Automate Gel Electrophoresis
5299	Performance Comparison of the Experion Automated Electrophoresis System and SDS-PAGE for Protein Analysis
5302	Performance Comparison of the Experion Automated Electrophoresis System and a Competing Automated System for Protein Analysis
5328	Experion Automated Electrophoresis System and the Experion Pro260 Analysis Kit: Accurate and Reproducible Protein Sizing and Quantitation in the Presence of High Salt Concentrations
5423	Protein Quantitation: A Comparative Analysis Using the Experion Automated Electrophoresis System, Bradford and Modified Lowry Assays, and SDS-PAGE
5453	Application of the Experion Automated Electrophoresis System to Glycoprotein Visualization and Analysis
5501	Rapid, Efficient Purification and Evaluation of His-Tagged Proteins
5506	Monitoring Development of Chromatographic Methods With the Experion Automated Electrophoresis System
5784	Comparison of Protein Quantitation Methods Using the Experion Automated Electrophoresis System

Appendix F: Ordering Information

Catalog #	Description
700-7000	Experion™ System, 100–240 V, for protein analysis, includes electrophoresis station, priming station, software, USB2 cable
701-7000	Experion System, 100–240 V, for protein analysis, includes electrophoresis station, priming station, software, USB2 cable, Experion Pro260 starter kit
700-7101	Experion Pro260 Analysis Kit for 10 Chips, includes 10 Pro260 chips, 1 cleaning chip, 3 x 520 µl Pro260 gel, 45 µl Pro260 stain, 60 µl Pro260 ladder (10–260 kD), 400 µl Pro260 sample buffer, 3 spin filters
700-7110	Experion Pro260 Starter Kit, includes 3 Experion chips, 1 cleaning chip, Experion reagents, spin filters, IgG protein standard, DTT, cleaning swabs (lint free), electrode cleaner, narrow bore polypropylene pipet tips, polypropylene 0.5 ml microcentrifuge tubes, DEPC-treated water (0.2 µm filtered)
700-7151	Experion Pro260 Chips, 10, plus 1 cleaning chip
700-7152	Experion Pro260 Reagents and Supplies, for 10 chips, includes 3 x 520 µl Pro260 gel, 45 µl Pro260 stain, 60 µl Pro260 ladder (10–260 kD), 400 µl Pro260 sample buffer, 3 spin filters
700-7256	Experion Protein Ladder, 60 µl
700-7251	Experion Cleaning Chips, 10
700-7254	Experion Spin Filters, 10
700-7252	Experion Electrode Cleaner, 250 ml
163-2091	ReadyPrep™ Proteomics Grade Water, 500 ml
161-0710	2-Mercaptoethanol, 25 ml
700-7051	Experion Validation Kit, 3 test chips, qualification procedures, dongle
700-7264	Cleaning Swabs, lint free, for electrode deep cleaning, 25
700-7031	Experion Priming Seals, 2



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